

Chapter 7

Centroids and Centers of Gravity

A **centroid** is the geometric center of a geometric object: a one-dimensional curve, a two-dimensional area or a three-dimensional volume. Centroids are useful for many situations in Statics and subsequent courses, including the analysis of distributed forces, beam bending, and shaft torsion.

Two related concepts are the **center of gravity**, which is the average location of an object's *weight*, and the **center of mass** which is the average location of an object's *mass*. In many engineering situations, the centroid, center of mass, and center of gravity are all coincident. Because of this, these three terms are often used interchangeably without regard to their precise meanings.

We consciously and subconsciously use centroids for many things in life and engineering, including:

Keeping your body's balance: Try standing up with your feet together and leaning your head and hips in front of your feet. You have just moved your body's center of gravity out of line with the support of your feet.

Computing the stability of objects in motion like cars, airplanes, and boats: By understanding how the center of gravity interacts with the accelerations caused by motion, we can compute safe speeds for sharp curves on a highway.

Designing the structural support to balance the structure's own weight and applied loadings on buildings, bridges, and dams: We design most large infrastructure not to move. To keep it from moving, we must understand how the structure's weight, people, vehicles, wind, earth pressure, and water pressure balance with the structural supports.

You probably have already developed a good intuition about centroids and centers of gravity based upon your life experience, and can roughly estimate their location when you look at an object or diagram. In this chapter you will learn to locate them precisely using two techniques: [integration 7.7](#) and the method of [composite parts 7.5](#).

7.1 Weighted Averages

You certainly know how to find the average of several numbers by adding them up and dividing by the number of values, so for example the average of the first four positive integers is

$$\frac{1 + 2 + 3 + 4}{4} = 2.5$$

More formally, if a is a set with n elements then the average, or **mean**, value is

$$\bar{a} = \frac{1}{n} \sum_{i=1}^n a_i = \frac{a_1 + a_2 + \cdots + a_n}{n}. \quad (7.1.1)$$

This average is also called the **arithmetic mean**. When calculating an arithmetic mean, each number is equally important when evaluating the average. The overbar symbol is often used to indicate that a quantity is a mean value.

In situations where some values are more important than others, we use a **weighted average**. A familiar example is your grade point average. Your GPA is calculated by weighting your grade for each class by the credits for that class, then dividing by the total credits you have taken. The credit values are called the **weighting factors**.

In general terms a weighted average is

$$\bar{a} = \frac{\sum a_i w_i}{\sum w_i} \quad (7.1.2)$$

Where a_i are the values we are averaging and w_i are the corresponding weighting factors. The weighting factors may be different for each item being averaged, so w_i is the weighting factor for value a_i . In this book we will not write the limits on the sums, and understand that the intent is always to sum over all the values. Notice that if the weighting factors are all identical, they can be factored out of the sums so the weighted average and the arithmetic mean will be the same.

Weighted averaging is used to find centroids, centers of gravity and centers of mass, the subject of this chapter. All three are *points* located at the “center” the object, but the meaning of “center” depends on the weighting factors. Area or volume are the factors used for centroids, weight for center of gravity, and mass for center of mass.

Example 7.1.1 Course Grades.

The mechanics syllabus says that there are two exams worth 25% each, homework is 10%, and the final is worth 40%. You have a 40 on the first exam, a 80 on the second exam, and your homework grade is 90.

What do you have to earn on the final exam to get a 70 in the class?

Solution. Your known grades and the weighting factors are

$$G_i = [40, 80, 90, FE]$$

$$w_i = [25\%, 25\%, 10\%, 40\%]$$

Find final exam score FE so that your average grade $\bar{G}$ is 70%.

$$\begin{aligned}\bar{G} &= \frac{\sum G_i w_i}{\sum w_i} \\ 70 &= \frac{(40 \times 0.25) + (80 \times 0.25) + (90 \times 0.1) + (FE \times 0.4)}{(0.25 + 0.25 + 0.1 + 0.4)} \\ FE &= \frac{70(1) - (10 + 20 + 9)}{0.4} = 77.5\end{aligned}$$

7.2 Center of Gravity

So far in this book we have always taken the weight of an object to act at a point at its center. This is the **center of gravity**: the point where all of an object's weight may be concentrated and still have the same *external* effect on the body. In this chapter we will learn to actually locate this point.

We will indicate the center of gravity with a circle with black and white quadrants, and call its coordinates $(\bar{x}, \bar{y})$ or $(\bar{x}, \bar{y}, \bar{z})$. This point represents the average location of all the particles which make up the body.

The center of gravity of a body is fixed with respect to the body, but the coordinates depend on the choice of coordinate system. For example, in [Figure 7.2.1](#) the center of gravity of the block is at its geometric center meaning that $\bar{x}$ and $\bar{y}$ are positive, but if the block is moved to the left of the y axis, or the coordinate system is translated to the right of the block, $\bar{x}$ would then become negative.

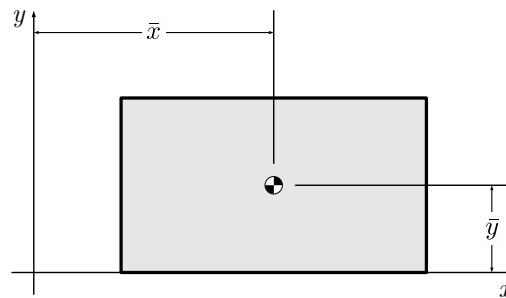


Figure 7.2.1 Location of the centroid, measured from the origin.

Lets explore the center of gravity of a familiar object. Take a pencil and try to balance it on your finger. How do you decide where to place it? You likely supported it roughly in the middle, then adjusted it until it balanced. You found the point where the moments of the weights on either side of your finger were in equilibrium.

Let's develop this balanced moment idea mathematically.

Assume that the two halves of the pencil have known weights acting at points 1 and 2. How could we replace the two weights with a single statically equivalent force? Recall from [Section 4.7](#) that statically equivalent systems produce the

same external effect on the object —the net force on the object, and the net moment about any point don't change. An upward force at this point will support the pencil without tipping.

To be equivalent, the total weight must equal the total weight of the parts. $W = W_1 + W_2$. Common sense also tells us that W will act somewhere between W_1 and W_2 .

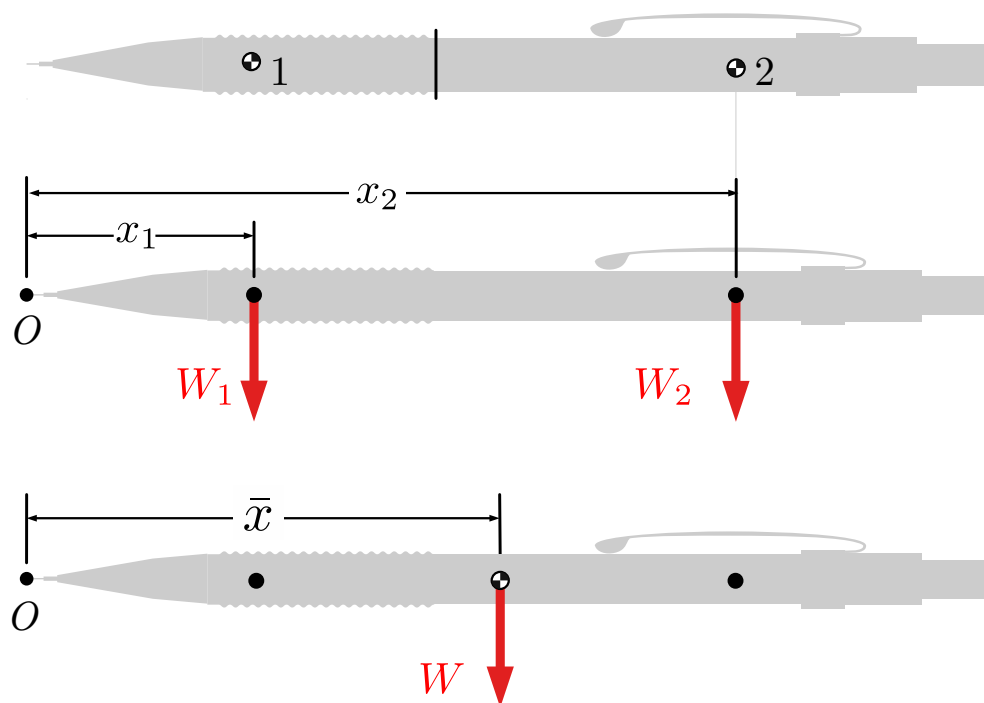


Figure 7.2.2 (top) Side view of a pencil representing each half as a particle. (middle) A force diagram showing the weights of the two particles. (bottom) An equivalent system consisting of a single weight acting at the pencil's center of gravity.

Next, let's do the mathematical equivalent of sliding your finger back and forth until a balance point is located. Pick any point O to be the origin, then calculate the total moment about O due to the two weights.

The sum of moments around point O can be written as:

$$\sum M_O = -x_1 W_1 - x_2 W_2$$

Notice that the moment of both forces are clockwise around point O , so the signs are negative according to the right-hand rule. We want a single equivalent force acting at the (unknown) center of gravity. Call the distance from the origin to the center of gravity $\bar{x}$.

$\bar{x}$ represents the mean distance of the weight, mass, or area depending on the context of the problem. We are evaluating weights in this problem, so $\bar{x}$ represents the distance from O to the center of gravity.

The sum of moments around point O for the equivalent system can be written as:

$$\sum M_O = -\bar{x}W$$

The moment of total weight W is also clockwise around point O , so the sign of moment will also be negative according to the right-hand rule. Since the two representation are equivalent we can equate them and solve for $\bar{x}$.

$$\begin{aligned} -\bar{x}W &= -x_1W_1 - x_2W_2 \\ \bar{x} &= \frac{x_1W_1 + x_2W_2}{W_1 + W_2} \end{aligned}$$

This result is exactly in the form of (7.1.2) where the value being averaged is distance x and the weighting factor is the weight of part W_i and the result is the mean distance $\bar{x}$.

The pencil was made up of two halves, but this equation can easily be extended n discrete parts. The resulting general definition of the centroidal coordinate $\bar{x}$ is:

$$\bar{x} = \frac{\sum \bar{x}_i W_i}{\sum W_i} \quad (7.2.1)$$

where:

W_i is the weight of part i ,

$\bar{x}_i$ is the x coordinate of the center of gravity of element i , and

$\sum$ is understood to mean “sum all parts” so there is no need to write $\sum_{i=1}^n$.

The numerator is the **first moment** of force which is literally a moment of force as we defined it in Chapter 3. The denominator is the sum of the weights of the pieces, which is the weight of the whole object. We will soon also see first moments of mass and first moments of area and in Chapter 10, we will introduce **second moments**, which are the integral of some quantity like area, multiplied by a distance *squared*.

We treated the pencil as a one-dimensional object, so this discussion focused on $\bar{x}$. There are similar formula for the other dimensions as well

$$\bar{x} = \frac{\sum \bar{x}_i W_i}{\sum W_i} \quad \bar{y} = \frac{\sum \bar{y}_i W_i}{\sum W_i} \quad \bar{z} = \frac{\sum \bar{z}_i W_i}{\sum W_i}. \quad (7.2.2)$$

In words, these equations say

$$\text{distance to CG} = \frac{\text{sum of first moments of weight}}{\text{total weight}}$$

They apply to any object which can be divided into discrete parts, and they produce the coordinates of the object’s center of gravity.

Question 7.2.3

Can you explain why the center of gravity of a symmetrical object will always fall on the axis of symmetry?

Answer. If the object is symmetrical, every subpart on the positive side of the axis of symmetry will be balanced by an identical part on the negative side. The first moment for the entire shape about the axis will sum to zero, meaning that

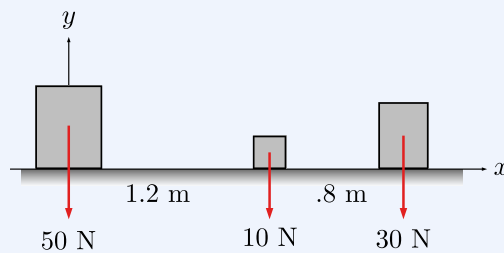
$$\bar{x} = \frac{\sum \bar{x}_i W_i}{\sum W_i} = \frac{0}{W} = 0.$$

In other words, the distance from the axis of symmetry of the shape to the centroid is zero.

Example 7.2.4 Simple Center of Gravity.

Three boxes are distributed along the x axis as shown.

- Find the distance from the origin to the center of gravity of the system.
- How would the center of gravity change if the origin was moved to the right-hand box?



Solution. The center of gravity of the system is the point where the total weight can be considered to act.

The total weight is $W = \sum W_i = 90$ N, and it acts at the point where it produces the same total moment as the individual weights. In other words, it is an *equivalent system*. The moment comparison may be taken about any point, so for convenience we choose the origin.

Let x_i represent the x -coordinates of the individual weights, and $\bar{x}$ represent the x -coordinate of the center of gravity. For both systems to be equivalent:

$$\begin{aligned} W \bar{x} &= \sum (W_i x_i) \\ (90 \text{ N}) \bar{x} &= (50 \text{ N})(0) + (10 \text{ N})(1.2 \text{ m}) + (30 \text{ N})(2.0 \text{ m}) \\ \bar{x} &= \frac{72 \text{ N}\cdot\text{m}}{90 \text{ N}} \\ &= 0.8 \text{ m} \end{aligned}$$

If the origin was moved to the right-hand box, the center of gravity would not change; however, the *coordinate* of the center of gravity would be

different. The distance to the center of gravity from the new origin would be:

$$\bar{x} = \frac{(10 \text{ N})(-0.8 \text{ m}) + (50 \text{ N})(-2.0 \text{ m})}{90 \text{ N}} = -1.2 \text{ m}$$

7.3 Center of Mass

The **center of mass** is the mean location of the mass of an object, and is related to the center of gravity by Newton's Second Law because

$$W = mg,$$

where g is the local strength of the gravitational field. In this course you may take $g = 9.81 \text{ m/s}^2$ in the SI system, or $g = 32.2 \text{ ft/s}^2$ in the US customary system as reasonable approximations for objects on the surface of the earth.

Substituting $m_i g_i = W_i$ in (7.2.2) gives the equations for the center of mass.

$$\bar{x} = \frac{\sum \bar{x}_i m_i g_i}{\sum m_i g_i} \quad \bar{y} = \frac{\sum \bar{y}_i m_i g_i}{\sum m_i g_i} \quad \bar{z} = \frac{\sum \bar{z}_i m_i g_i}{\sum m_i g_i}. \quad (7.3.1)$$

By our assumption that g is constant on the surface of the earth, g_i can be factored out of the sums and drops out of the equation completely.

$$\bar{x} = \frac{\sum \bar{x}_i m_i}{\sum m_i} \quad \bar{y} = \frac{\sum \bar{y}_i m_i}{\sum m_i} \quad \bar{z} = \frac{\sum \bar{z}_i m_i}{\sum m_i}. \quad (7.3.2)$$

These equations give the coordinates of the center of mass. The numerator contains the **first moment of mass**, and the denominator contains the total mass of the object. As long as the assumption that g is constant is valid, the center of mass and the center of gravity are identical points and the two terms may be used interchangeably.

7.4 Centroids

Key Questions

- What is the difference between a centroid, center of gravity and a center of mass?
- When will the centroid, center of gravity and center of mass refer to the same point?
- Why do the equations for the center of gravity, mass, volume, and area all have the same structure?

A centroid is a weighted average like the center of gravity, but weighted with a geometric property like area or volume, and not a physical property like weight or mass. This means that centroids are properties of pure shapes, not physical objects. They represent the coordinates of the “middle” of the shape.

The defining equations for centroids are similar to the equations for [Centers of Gravity \(7.2.2\)](#) but with *volume* used as the weighting factor for three-dimensional shapes

$$\bar{x} = \frac{\sum \bar{x}_i V_i}{\sum V_i} \quad \bar{y} = \frac{\sum \bar{y}_i V_i}{\sum V_i} \quad \bar{z} = \frac{\sum \bar{z}_i V_i}{\sum V_i}, \quad (7.4.1)$$

and *area* for two-dimensional shapes

$$\bar{x} = \frac{\sum \bar{x}_i A_i}{\sum A_i} \quad \bar{y} = \frac{\sum \bar{y}_i A_i}{\sum A_i}. \quad (7.4.2)$$

We will see how to use these equations on complex shapes later in this chapter, but centroids of some simple shapes can be easily found using symmetry.

If the shape has an axis of symmetry, every point on one side of the axis is mirrored by another point equidistant on the other side. One has a positive distance from the axis, and the other is the same distance away in the negative direction. These two points will add to zero the numerator, as will every other point making up the shape, and the first moment will be zero. This means that the centroid must lie along the line of symmetry if there is one. If a shape has multiple symmetry lines, then the centroid must exist at their intersection.

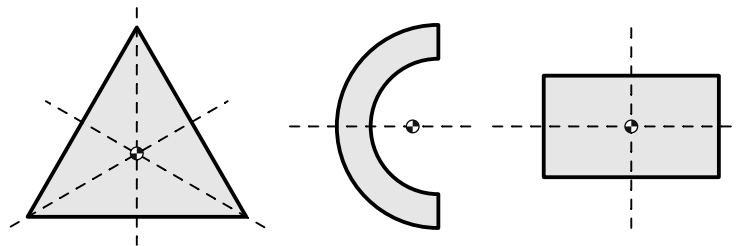
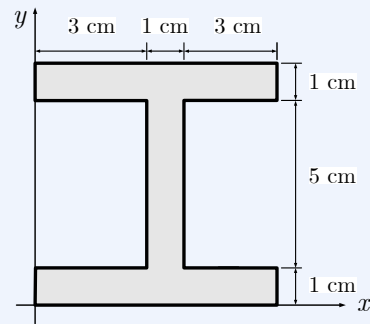


Figure 7.4.1 Centroids lie upon axes of symmetry.

Since rectangles, circles, cubes, spheres, etc. have multiple lines of symmetry, their centroids must be exactly in the center as we would expect.

Question 7.4.2

What are the coordinates of the centroid of the I beam section shown?



Answer.

$$\bar{x} = \bar{y} = 3.5 \text{ cm}$$

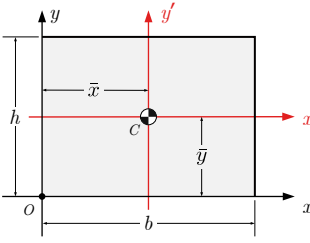
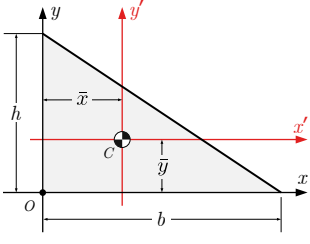
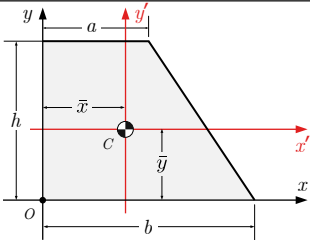
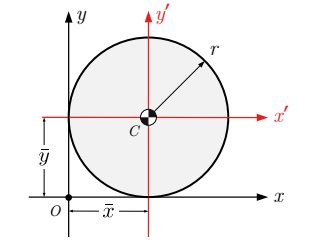
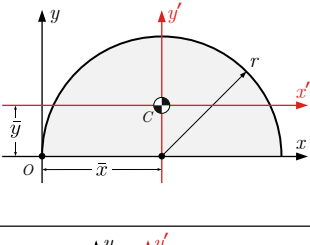
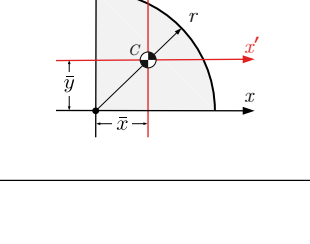
Solution. The cross section is symmetrical about both a vertical and horizontal centerline. The centroid is at the intersection, in the middle. The coordinates are measured from the origin, in the bottom left of the diagram.

$$\bar{x} = \bar{y} = 3.5 \text{ cm}$$

7.4.1 Properties of Common Shapes

We will learn how to find centroids of other shapes in [Section 7.7](#) using integration, but in the mean time several common shapes are recorded in the table below. This information in this table will be needed in the next section.

Table 7.4.3 Centroids of Common Shapes

Shape	Area	$\bar{x}$	$\bar{y}$
	$A = bh$	$b/2$	$h/2$
	$\frac{bh}{2}$	$b/3$	$h/3$
	$\frac{(a+b)h}{2}$	$\frac{a^2 + ab + b^2}{3(a+b)}$	$\frac{h(2a+b)}{3(a+b)}$
	πr^2	r	r
	$\frac{\pi r^2}{2}$	r	$\frac{4r}{3\pi} \approx 0.424r$
	$\frac{\pi r^2}{4}$	$\frac{4r}{3\pi}$	$\frac{4r}{3\pi}$

Note 7.4.4

In this table, all centroids are measured from the indicated origin. You must make the appropriate adjustments when the origin of your coordinate system is located elsewhere.

7.4.2 Relations between Centroids and Center of Gravity

The equations we have been discussing (7.2.2), (7.3.1), (7.4.1) and (7.4.2) are all variations on the general weighted average formula (7.1.2).

$$\bar{a} = \frac{\sum a_i w_i}{\sum w_i}$$

Here a_i represents the distance in one of the coordinate directions such as x , $\bar{a}$ is the mean distance in the a direction to the ‘middle’ of the whole object, and w is the weighting factor. The only difference between them is the choice of weighting factor. For center of gravity, the weighting factor is the weight, for center of mass, it is the mass, for three dimensional centroids it is the volume, and for two dimensional centroids it is area.

To understand how these equations relate to one another consider a plate with a cross-sectional area A , divided into n pieces with volume V_i .

The weight of part i is the product of its specific weight and volume.

$$W_i = \gamma V_i = \rho_i g_i A_i t_i$$

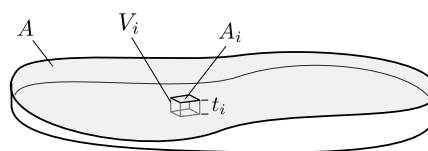


Figure 7.4.5 Plate with variable thickness t , divided into many volume elements V_i .

In the most general case, all of these terms might depend on the position of the part, but if any are constant they can be factored out and simplify the formulas.

For a homogeneous flat plate with uniform thickness, like a piece of plywood, the density, thickness and g are all constant so

$$W_i = \rho g t A_i$$

$$\begin{aligned} \bar{x} &= \frac{\sum \bar{x}_i W_i}{\sum W_i} & \bar{y} &= \frac{\sum \bar{y}_i W_i}{\sum W_i} & \bar{z} &= \frac{\sum \bar{z}_i W_i}{\sum W_i} \\ \bar{x} &= \frac{\rho g t \sum \bar{x}_i A_i}{\rho g t \sum A_i} & \bar{y} &= \frac{\rho g t \sum \bar{y}_i A_i}{\rho g t \sum A_i} & \bar{z} &= \frac{\rho g t \sum \bar{z}_i A_i}{\rho g t \sum A_i} \end{aligned}$$

The two dimensional centroid equations are sufficient to find the center of gravity of a three dimensional object.

7.5 Centroids using Composite Parts

Key Questions

- How do you calculate the center of gravity of a system of separate objects?
- Where do the equations for the shapes in areas and centroids table come from?
- When finding the centroid, what do you do with a cut-out area of a composite part?
- Does it matter whether a distance to the centroid of a part is positive or negative from the axis system?

In this section we will discuss how to find centroids of two-dimensional shapes by first dividing them into pieces with known properties, and then combining the pieces to find the centroid of the original shape. This method will work when the geometric properties of all the sub-shapes are known or can be easily determined. If the shape can't be decomposed this way, perhaps because it has a curved boundary, you will need to use integration to find the centroid. Integration will be covered in [Section 7.7](#).

For convenience, the properties of several common shapes can be found [here](#).

7.5.1 Composite Parts Method

The equations we will use for this approach are

$$\bar{x} = \frac{\sum \bar{x}_i A_i}{\sum A_i} \quad \bar{y} = \frac{\sum \bar{y}_i A_i}{\sum A_i} \quad (7.5.1)$$

where,

$\bar{x}$, and $\bar{y}$ are the coordinates of the centroid of the entire shape.

A_i is the area of composite part i .

$\bar{x}_i$, and $\bar{y}_i$ are the coordinates of the centroid of composite part i .

The steps to finding a centroid using the composite parts method are:

1. Break the overall shape into simpler parts.
2. Collect the areas and centroid coordinates, and
3. Apply (7.5.1) to combine to find the coordinates of the centroid of the original shape.

As a simple example, consider the L-shaped area shown, which has been divided into two rectangles. The areas of the rectangles are

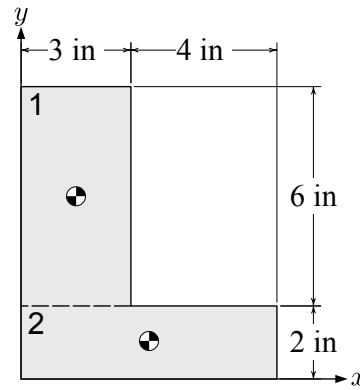
$$A_1 = 18 \text{ in}^2, A_2 = 14 \text{ in}^2$$

The origin is located at the lower left, so the coordinates of the centroids of the two rectangles are

$$\bar{x}_1 = 1.5 \text{ in}, \bar{y}_1 = 5 \text{ in}, \quad \bar{x}_2 = 3.5 \text{ in}, \bar{y}_2 = 1 \text{ in}$$

The centroid of the whole shape is found by applying (7.5.1)

$$\begin{aligned} \bar{x} &= \frac{\sum \bar{x}_i A_i}{\sum A_i} & \bar{y} &= \frac{\sum \bar{y}_i A_i}{\sum A_i} \\ \bar{x} &= \frac{(1.5)(18) + (3.5)(14)}{18 + 14} & \bar{y} &= \frac{\sum (5)(18) + (1)(14)}{18 + 14} \\ \bar{x} &= 2.375 \text{ in} & \bar{y} &= 3.25 \text{ in} \end{aligned}$$



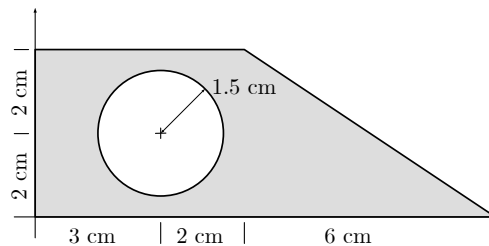
For more complex shapes, the usual practice is to set up a table to organize the information needed to calculate the centroid, as we will now show. The process can be broken into three steps.

1. *Break the overall shape into simpler parts.*

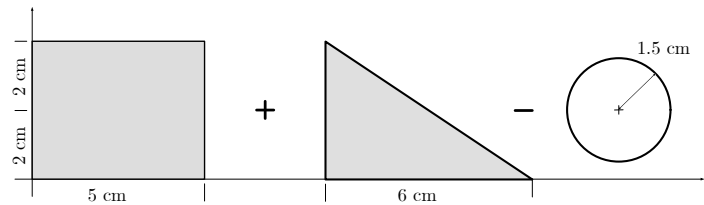
We begin with a sketch of the shape and establish a coordinate system. It is critical that all measurements are made from a common origin, and the results will be measured from this origin as well. A careful choice of origin can simplify the problem, so give it some thought.

Then divide the shape into several simpler shapes. The sub-shapes may include holes, which are treated as negative areas. You must know how to calculate the area and locate the centroid of any sub-shape you use.

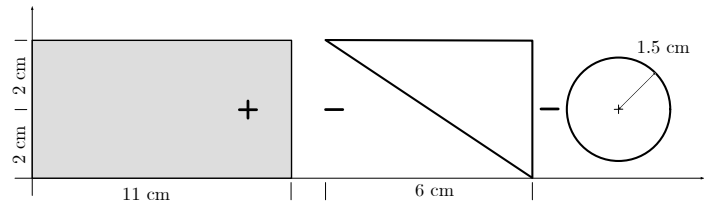
Consider the complex shape below.



There are often several ways to divide a shape, but it's best to use as few parts as possible to minimize your computations and opportunities for error. For example, you could choose to break this shape into either a 5 cm × 4 cm rectangle, a 6 cm × 4 cm right triangle, and an $r = 1.5$ cm circular hole,



or a large 11 cm $\times$ 4 cm rectangle, an $r = 1.5$ cm circular hole, and a 6 cm $\times$ 4 cm right triangle subtracted from the large rectangle.



Both options will give the same results, and in this case there is no particular advantage to one choice over the other. However, it would be silly and unnecessary to break this into more than three parts, and it would not be a good idea to divide this into a trapezoid minus a hole, unless you know geometric properties of a trapezoid, which are not available in [Subsection 7.4.1](#). Be sure your sub-shapes don't overlap and don't get counted more than once.

2. *Collect the areas and centroid coordinates.*

Once the complex shape has been divided into parts, the next step is to determine the area and centroidal coordinates for each part. You can use the properties in [Subsection 7.4.1](#) for rectangles, triangles, circles, semi-circles and quarter circles but you will need to use integration if other shapes are involved. Any holes or removed shapes should be treated as negative areas.

Record the information you gather in a table like the one below. The table should include a row containing column headings and units, one row for each part, and a summary row. The first column identifies the part — by number or sketch, the second contains the areas, and the third and fourth contains the centroidal coordinates of the parts.

Part	A_i [cm ²]	$\bar{x}_i$ [cm]	$\bar{y}_i$ [cm]	$A_i\bar{x}_i$ [cm ³]	$A_i\bar{y}_i$ [cm ³]
1	20	2.5	2	50	40
2	12	7	4/3	84	16
3	-2.25π	3	2	-6.75π	-4.5π
Σ	24.93	—	—	112.8	41.86

The last two columns of the table contain the first moments of area $Q_x = A_i\bar{y}_i$ and $Q_y = A_i\bar{x}_i$, and are easily filled in by multiplying the values in

columns two to four. Be sure to attend to positive and negative signs when multiplying. Note that the moment of area with respect to the x axis uses the distance from the x axis, which is $\bar{y}_i$, and vice-versa.

The final row of the table are total values, calculated by summing the entries for A_i , Q_x and Q_y , so for example the total area of the shape is

$$A = \sum A_i = A_1 + A_2 + A_3 \dots$$

Don't sum columns three or four, since $\sum \bar{x}_i$ and $\sum \bar{y}_i$ are meaningless.

3. *Combine the pieces to find the overall centroid.*

After you have filled in the whole table, you can find the coordinates of the centroid by applying (7.5.1) with the summary values from the last row.

$$\begin{aligned}\bar{x} &= \frac{Q_y}{A} &= \frac{112.8}{24.93} = 4.52 \text{ cm} \\ \bar{y} &= \frac{Q_x}{A} &= \frac{41.86}{24.93} = 1.692 \text{ cm}\end{aligned}$$

Finally, plot the centroid $(\bar{x}, \bar{y})$ on the diagram. If you have made a calculation error it will usually be obvious, because the centroid location won't "feel right."

The next interactive shows a composite shape consisting of a large rectangle with a smaller rectangle subtracted. You can change the location and size of the rectangles by moving the red points. Use this to visualize how the centroids of the whole is related to the centroid of the parts. Note that for objects divided into two pieces, the centroid of the whole always falls on the line connecting the centroids of the parts.

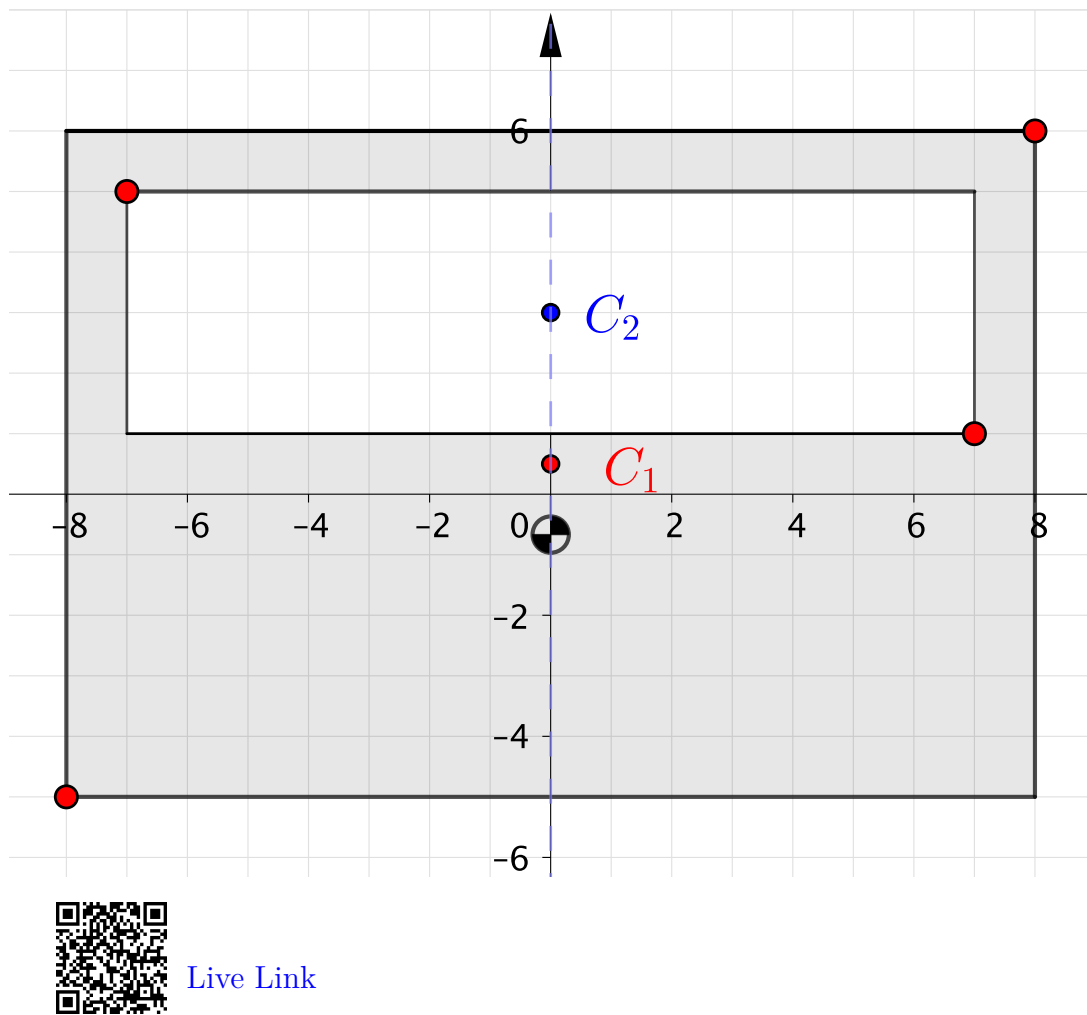


Figure 7.5.1 Centroid of Composite Rectangles

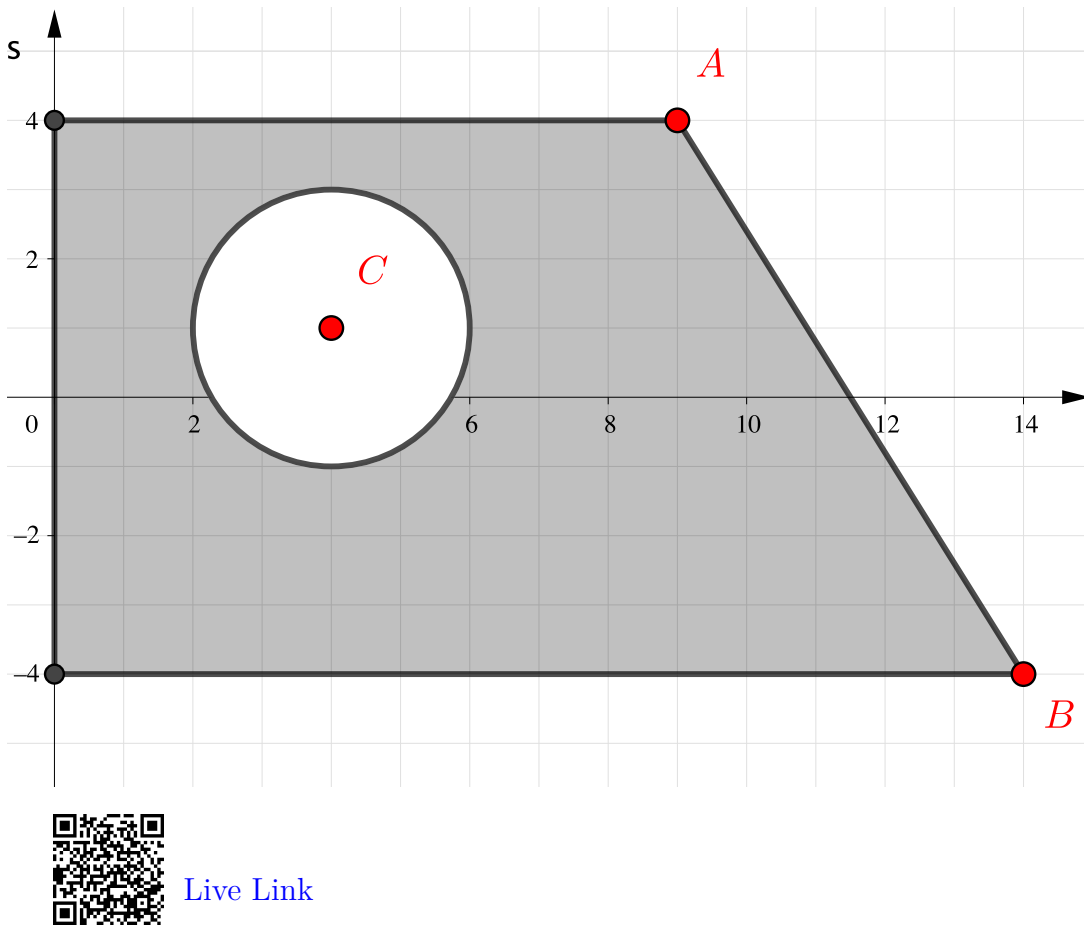


Figure 7.5.2 Centroid of a body consisting of a rectangle, triangle and a circular hole.

7.5.2 Centroids of 3D objects

Key Questions

- How do you divide a composite solid into parts and compute the volume/mass and centroidal distances of each part?
- What is the technique to compute the overall center of volume/mass for a composite solid?

The centroid of a three-dimensional volume is found similarly to two-dimensional centroids, but with volume used instead of area for the weighting factor. The centroid of a volume and the center of mass or gravity for a homogenous solid are identical.

$$\bar{x} = \frac{\sum \bar{x}_i V_i}{\sum V_i} \quad \bar{y} = \frac{\sum \bar{y}_i V_i}{\sum V_i} \quad \bar{z} = \frac{\sum \bar{z}_i V_i}{\sum V_i}$$

Where,

$\bar{x}$, $\bar{y}$, and $\bar{z}$ are the coordinates of the centroid of the overall volume. V_i is the volume of composite part i .

$\bar{x}_i$, $\bar{y}_i$, and $\bar{z}_i$ is the coordinates of the centroid of composite part i .

Many three-dimensional shapes are just prismatic extrusions of the shapes. The volume of a prism is the product of the cross-sectional area and the length of the prism and is easily calculated. For example, the volume of a circular cylinder with radius r and length l is $V = \pi r^2 l$.

If the density varies for each part of a composite solid, we can find the center of mass by dividing the first moment of mass by the total mass. You can also compute the center of gravity by replacing the mass terms in the equations below with weight terms.

$$\bar{x} = \frac{\sum \bar{x}_i m_i}{\sum m_i} \quad \bar{y} = \frac{\sum \bar{y}_i m_i}{\sum m_i} \quad \bar{z} = \frac{\sum \bar{z}_i m_i}{\sum m_i}$$

Here m_i is the mass of composite part i .

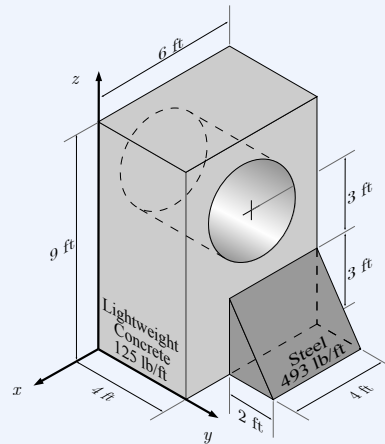
You must always use the same weighting factor (area, volume, mass, weight, etc) in both the numerator and denominator of the center of area/volume/mass/weight equations.

Example 7.5.3 3D Center of Mass.

A composite solid consists of a rectangular block of lightweight concrete and a triangular wedge of steel with dimensions as shown. The rectangular block has a 2 ft radius circular hole, centered and drilled through its full depth, perpendicular to the front and back faces.

Assume $\gamma_C = 125 \text{ lb/ft}^3$, and $\gamma_S = 493 \text{ lb/ft}^3$.

Find the center of mass of this composite solid.



Solution.

Table 7.5.4

Part	V_i [ft ³]	γ [lb/ft ³]	W_i [lb]	$\bar{x}_i$ [ft]	$\bar{y}_i$ [ft]	$\bar{z}_i$ [ft]	$W_i \bar{x}_i$ [ft·lb]	$W_i \bar{y}_i$ [ft·lb]	$W_i \bar{z}_i$ [ft·lb]
block	216	125	27000	-3	2	4.5	-81000	54000	121500
hole	-50.27	125	-6283	-3	2	6	18850	-12566	-37699
wedge	12	493	5916	-4	4.67	1	-23664	27608	5916
			26633				-85814	69042	89717

$$\begin{aligned}\bar{x} &= \frac{\sum W_i \bar{x}_i}{\sum W_i} = \frac{-85814 \text{ ft}\cdot\text{lb}}{26633 \text{ lb}} = -3.22 \text{ ft} \\ \bar{y} &= \frac{\sum W_i \bar{y}_i}{\sum W_i} = \frac{69042 \text{ ft}\cdot\text{lb}}{26633 \text{ lb}} = 2.59 \text{ ft} \\ \bar{z} &= \frac{\sum W_i \bar{z}_i}{\sum W_i} = \frac{89717 \text{ ft}\cdot\text{lb}}{26633 \text{ lb}} = 3.37 \text{ ft}\end{aligned}$$

We have actually found the coordinates of the center of gravity, but since g is constant they are also coordinates of the center of mass.

7.6 Average Value of a Function

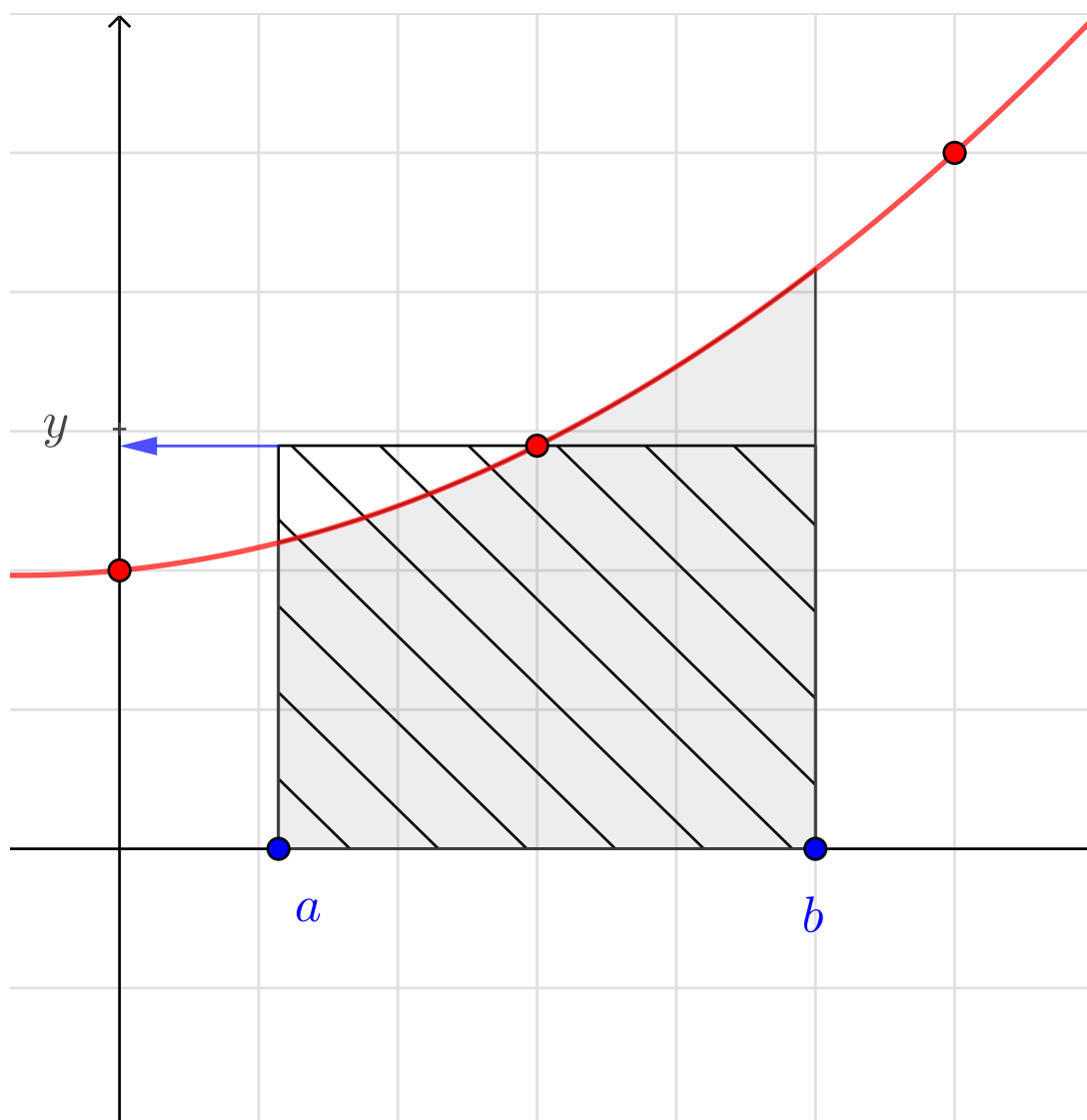
The weighted average technique discussed in (7.1.2) are a fine for averaging several discrete values, but what do we do if we need to find the average of an infinite number of values or values which change continuously?

Consider a function $y = f(x)$ over some interval from a to b . How can we find the average value $\bar{y}$ of the function over that interval? To understand what is meant by the average value of a function, look at the interactive below. There, the function $f(x)$ is the red curve, and which you can change if you like. The blue dots a and b mark the beginning and end of the interval and are also adjustable.

The area under this curve between a and b is shaded with gray, and we can find it using a definite integral

$$\int_a^b f(x) dx$$

The blue hatched rectangle has the same area as the gray shaded region, and because the areas are the same, the height of the rectangle $\bar{y}$, is the average value of $f(x)$.



[Live Link](#)

Figure 7.6.1 The average value of a function between a and b .

With this in mind, we can calculate the average value of $f(x)$ by equating the area under the curve with the area of the rectangle and solving for $\bar{y}$.

$$\int_a^b f(x) \, dx = \bar{y}(b - a)$$

$$\bar{y} = \frac{\int_a^b f(x) \, dx}{(b - a)}$$

$$\bar{y} = \frac{\int_a^b f(x) \, dx}{\int_a^b 1 \, dx}$$

$$\text{and since } \int_a^b 1 \, dx = (b - a)$$

This is a weighted average like (7.1.2) but instead of summing n discrete values, we integrate of an infinite number of infinitesimal values. $f(x)$ is the value being averaged and the weighting function is dx .

This approach is true for any choice of weighting function. To find $\bar{x}$ for a two dimensional area, the value to be averaged is x and the weighting function is dA , so replacing dx with dA and x_i with x ,

$$\bar{x} = \frac{\sum \bar{x}_i A_i}{\sum A_i} \quad \Rightarrow \quad \bar{x} = \frac{\int x dA}{\int dA}$$

In other words, to transform a discrete summation to an equivalent continuous integral form you:

1. Replace the summation with integration, $\Sigma \Rightarrow \int$.
2. Replace the discrete weighting factor with the corresponding differential element,

$$\begin{cases} A_i & \Rightarrow dA \\ V_i & \Rightarrow dV \\ W_i & \Rightarrow dW \end{cases} \quad \text{etc.}$$

3. Rename the value being averaged to eliminate the index i . We often use el as a subscript when referring to a differential element.

The two-dimensional centroid equations (7.5.1) become,

$$\bar{x} = \frac{\sum \bar{x}_i A_i}{\sum A_i} \Rightarrow \frac{\int \bar{x}_{el} dA}{\int dA} \quad \bar{y} = \frac{\sum \bar{y}_i A_i}{\sum A_i} \Rightarrow \frac{\int \bar{y}_{el} dA}{\int dA}$$

, and in the same way the center of gravity equations become

$$\bar{x} = \frac{\int \bar{x}_{el} dW}{\int dW} \quad \bar{y} = \frac{\int \bar{y}_{el} dW}{\int dW} \quad \bar{z} = \frac{\int \bar{z}_{el} dW}{\int dW}$$

Question 7.6.2

How far is it from the earth to the sun?

Answer. 92,958,412 miles

Solution. Siri says that “The average distance from the earth to the sun is 92,958,412 miles.”

That’s a pretty exact answer. What does it mean, exactly? From what point on the earth to what point on the sun?

If the earth and sun were perfect spheres, we could use the distance between their centroids. With more information about shape and density, we could find their centers of mass and measure between those points.

The bigger problem is that this distance changes continuously as the earth revolves around the sun. How can we find an average value for something which is continuously changing?

We need to use the methods described here, integrating the distance as a function of time over the course of a year.

7.7 Centroids using Integration

Key Questions

- How do you find the centroid of an area using integration?
- What is a differential quantity?
- Why are double integrals required for square dA elements and single integrals required for rectangular dA elements?

In this section we will use the integral form of (7.4.2) to find the centroids of non-homogenous objects or shapes with curved boundaries.

$$\bar{x} = \frac{\int \bar{x}_{el} dA}{\int dA} \quad \bar{y} = \frac{\int \bar{y}_{el} dA}{\int dA} \quad \bar{z} = \frac{\int \bar{z}_{el} dA}{\int dA} \quad (7.7.1)$$

With the integral equations we are mathematically breaking up a shape into an infinite number of infinitesimally small pieces and adding them together by integrating. This powerful method is conceptually identical to the discrete sums we introduced first.

7.7.1 Integration Process

Determining the centroid of a area using integration involves finding weighted average values $\bar{x}$ and $\bar{y}$, by evaluating these three integrals,

$$A = \int dA, \quad Q_x = \int \bar{y}_{el} dA \quad Q_y = \int \bar{x}_{el} dA \quad (7.7.2)$$

, where

- dA is a differential bit of area called the *element*.
- A is the *total area* enclosed by the shape, and is found by evaluating the first integral.
- $\bar{x}_{el}$ and $\bar{y}_{el}$ are the coordinates of the *centroid of the element*. These are frequently functions of x or y , not constant values.

- Q_x and Q_y are the *First moments of Area* with respect to the x and y axis.

The procedure for finding centroids with integration can be broken into three steps:

1. *Set up the integrals.*

Usually this is the hardest step.

You should always begin by drawing a sketch of the problem and reviewing the given information.

You will need to understand the boundaries of the shape, which may be lines or functions. You may need to know some math facts, like the definition of slope, or the equation of a line or parabola. A bounding function may be given as a function of x , but you want it as a function of y , or vice-versa or it may have a constant which you will need to determine.

You will need to choose an element of area dA . There are several choices available, including vertical strips, horizontal strips, or square elements; or in polar coordinates, rings, wedges or squares. There really is no right or wrong choice; they will all work, but one may make the integration easier than another. The best choice depends on the nature of the problem, and it takes some experience to predict which it will be.

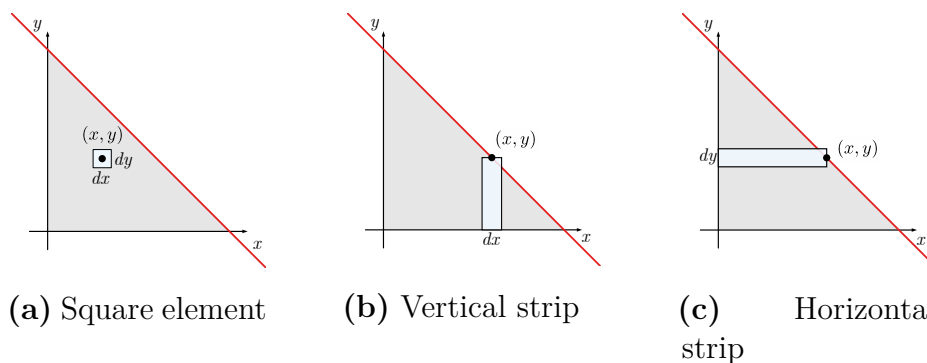


Figure 7.7.1 Differential Elements of Area

The two most common choices for differential elements are:

- *Square elements and double integrals.*

If you choose an infinitesimal square element $dA = dx \, dy$, you must integrate twice, over x and over y between the appropriate integration limits. The position of the element typically designated (x, y) .

- *Rectangular elements and single integrals.*

If you choose rectangular strips you eliminate the need to integrate twice. You may select a vertical element with a different width dx , and a height extending from the lower to the upper bound, or a horizontal strip with a differential height dy and a width extending from the left to the right boundaries. Either way, you only integrate once to cover the enclosed area.

When finding the area enclosed by a single function $y = f(x)$, and the x and y axes (x, y) represents a point on the function and $dA = y \, dx$ for vertical strips, or $dA = x \, dy$ for horizontal strips.

You must find expressions for the area dA and centroid of the element $(\bar{x}_{el}, \bar{y}_{el})$ in terms of the bounding functions. This is how we turn an integral over an area into a definite integral which can be integrated.

When you have established all these items, you can substitute them into (7.7.2) and proceed to the integration step.

2. Solve the integrals.

This step is pure mathematics.

Here are some tips if you are doing integration “by hand”. Be neat, work carefully, and check your work as you go along. Use proper mathematics notation: don’t lose the differential dx or dy before the integration step, and don’t include it afterwards. Don’t forget to use equals signs between steps. Simplify as you go and don’t substitute numbers or other constants too soon. Pay attention to units: Area A should have units of $[\text{length}]^2$ and the first moments of area Q_x and Q_y should have units of $[\text{length}]^3$. If your units aren’t consistent, then you have made a mistake.

3. Evaluate the centroid.

After you have evaluated the integrals you will have expressions or values for A , Q_x , and Q_y . All that remains is to substitute these into the defining equations for $\bar{x}$ and $\bar{y}$ and simplify. Notice the Q_x goes into the $\bar{y}$ equation, and vice-versa.

$$\bar{x} = \frac{Q_y}{A} \qquad \bar{y} = \frac{Q_x}{A}$$

Finally, plot the centroid at $(\bar{x}, \bar{y})$ on your sketch and decide if your answer makes sense for area.

Thinking Deeper 7.7.2 What is dA ?

dA is just an area, but an extremely tiny one!

It’s an example of an **differential quantity**—also called an **infinitesimal**. A differential quantity is a value which is as close to zero as it can possibly be without actually being zero. You can think of its value as $\frac{1}{\infty}$. Integration is the process of adding up an infinite number of infinitesimal quantities.

Some other differential quantities we will see in statics are dx , dy and dz , which are infinitesimal increments of distance; dV , which is a differential volume; dW , a differential weight; dm , a differential mass, and so on.

Any product involving a differential quantity is itself a differential quantity,

so if the area of a vertical strip is given by $dA = y \, dx$ then, even though height y is a real number, the area is a differential because dx is differential. If you like, you can pronounce the d as “the little bit of” so $dA = y \, dx$ reads “The little bit of area is the height y times a little bit of x .” and $A = \int dA$ reads “The total area is the sum of the little bits of area.”

7.7.2 Area of a General Spandrel

In this section we will use the integration process describe above to calculate the area of the general spandrel shown in [Figure 7.7.3](#). A **spandrel** is the area between a curve and a rectangular frame. This is a general spandrel because the curve is defined by the function $y = kx^n$, where n is not specified. If $n = 0$ the function is constant, if $n = 1$ then it is a straight line, $n = 2$ it's a parabola, etc. You can change the slider in the interactive below to see the effect of different values of n .

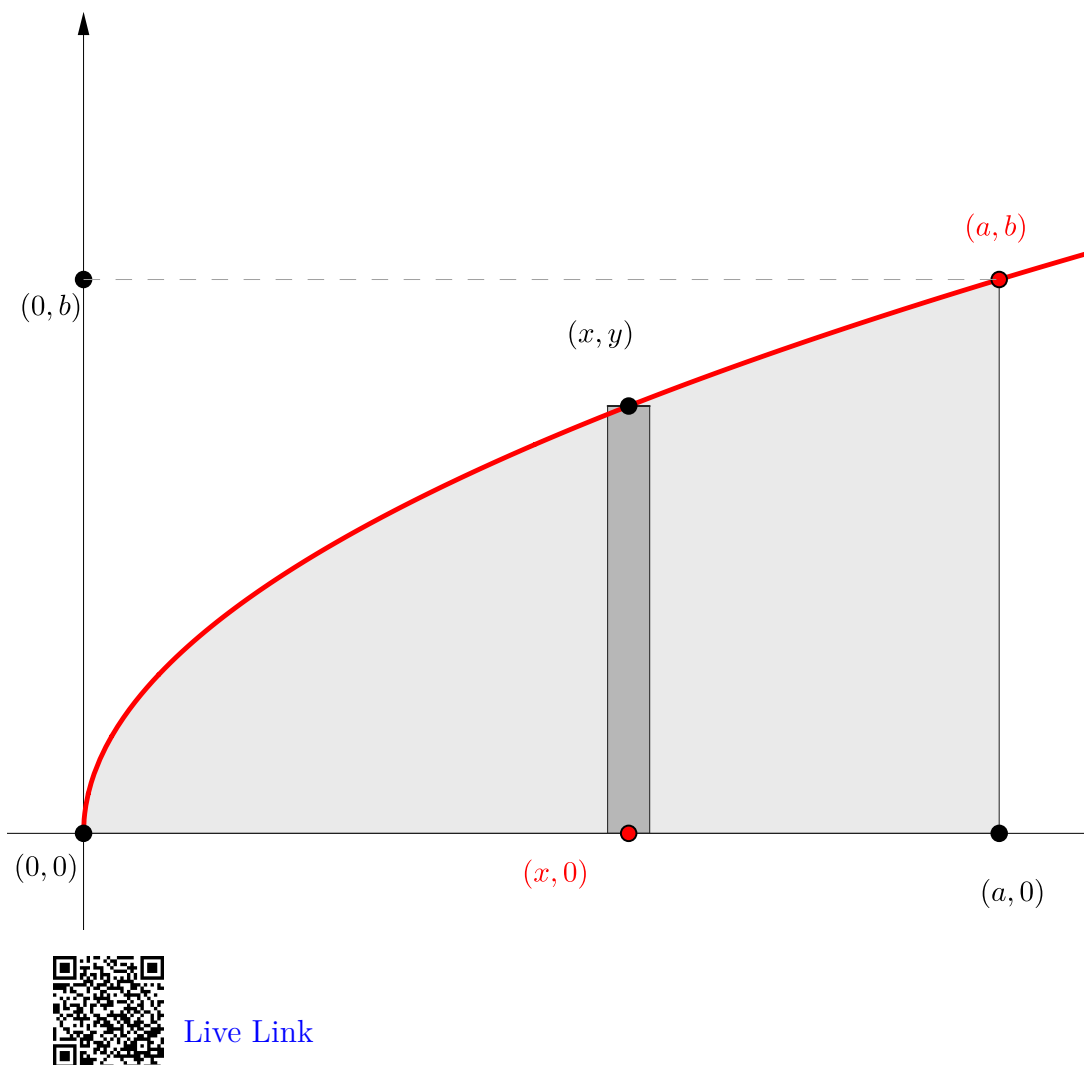


Figure 7.7.3 A general spandrel of the form $y = kx^n$.

Begin by identifying the bounding functions. From the diagram, we see that the boundaries are the function, the x axis and, the vertical line $x = b$. The function $y = kx^n$ has a constant k which has not been specified, but which is not arbitrary. The diagram indicates that the function passes through the origin and point (a, b) , and there is only one value of k which will cause this. We can find k by substituting a and b into the function for x and y then solving for it.

$$\begin{aligned} y &= kx^n \\ b &= ka^n \\ k &= \frac{b}{a^n} \end{aligned}$$

Next, choose a differential area. For this problem a vertical strip works well. A vertical strip has a width dx , and extends from the bottom boundary to the

top boundary. Any point on the curve is (x, y) and a point directly below it on the x axis is $(x, 0)$. This means that the height of the strip is $(y - 0) = y$ and the area of the strip is (base $\times$ height), so

$$dA = y \, dx.$$

The limits on the integral are from $x = 0$ on the left to $x = a$ on the right since we are integrating with respect to x .

With these details established, the next step is to set up and evaluate the integral $A = \int dA = \int_0^a y \, dx$. This is the familiar formula from calculus for the area under a curve. Proceeding with the integration

$$\begin{aligned} A &= \int dA && (dA = y \, dx) \\ &= \int_0^a y \, dx && (y = kx^n) \\ &= \int_0^a kx^n \, dx && (\text{integrate}) \\ &= k \left. \frac{x^{n+1}}{n+1} \right|_0^a && (\text{evaluate limits}) \\ &= k \frac{a^{n+1}}{n+1} && \left(k = \frac{b}{a^n} \right) \\ &= \frac{b}{a^n} \frac{a^{n+1}}{n+1} && (\text{simplify}) \\ A &= \frac{ab}{n+1} && (\text{result}) \end{aligned}$$

This result is not a number, but a general formula for the area under a curve in terms of a , b , and n . Explore with the interactive, and notice for instance that when $n = 0$, the shape is a rectangle and $A = ab$; when $n = 1$ the shape is a triangle and the $A = ab/2$; when $n = 2$ the shape is a parabola and $A = ab/3$ etc. This single formula gives the equation for the area under a whole family of curves.

Thinking Deeper 7.7.4 Which is better, horizontal or vertical elements?

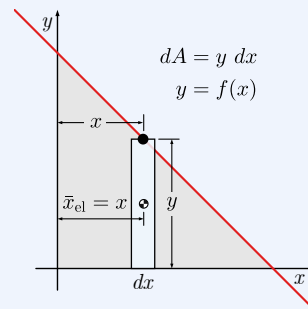
Recall that the first moment of area $Q_x = \int \bar{x}_{\text{el}} \, dA$ is the distance weighted area as measured from a desired axis. The distance term $\bar{x}_{\text{el}}$ is the distance from the desired axis to the centroid of each differential element of area, dA .

If you're using a single integral with a *vertical* element dA

$$dA = \underbrace{f(x)}_{\text{height}} \underbrace{(dx)}_{\text{base}} = y \, dx$$

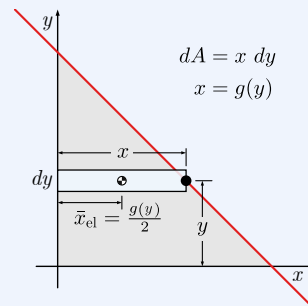
and the horizontal distance from the y axis to the centroid of dA would simply be

$$\bar{x}_{\text{el}} = x.$$



It is also possible to find $\bar{x}$ using a horizontal element but the computations are a bit more challenging. First the equation for dA changes to

$$dA = \underbrace{g(y)}_{\text{base}} \underbrace{(dy)}_{\text{height}} = x \, dy.$$



Additionally $\bar{x}_{\text{el}}$, must measure the distance to the centroid of the horizontal element. For these horizontal strips,

$$\bar{x}_{\text{el}} = \frac{x}{2}.$$

Similarly

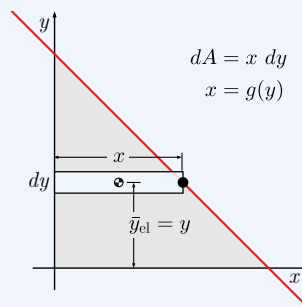
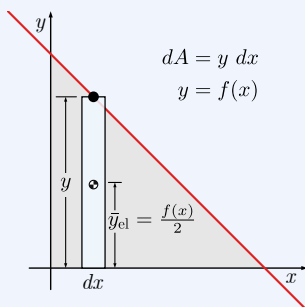
$$\bar{y}_{\text{el}} = y$$

for horizontal elements, but

$$\bar{y}_{\text{el}} = \frac{y}{2}$$

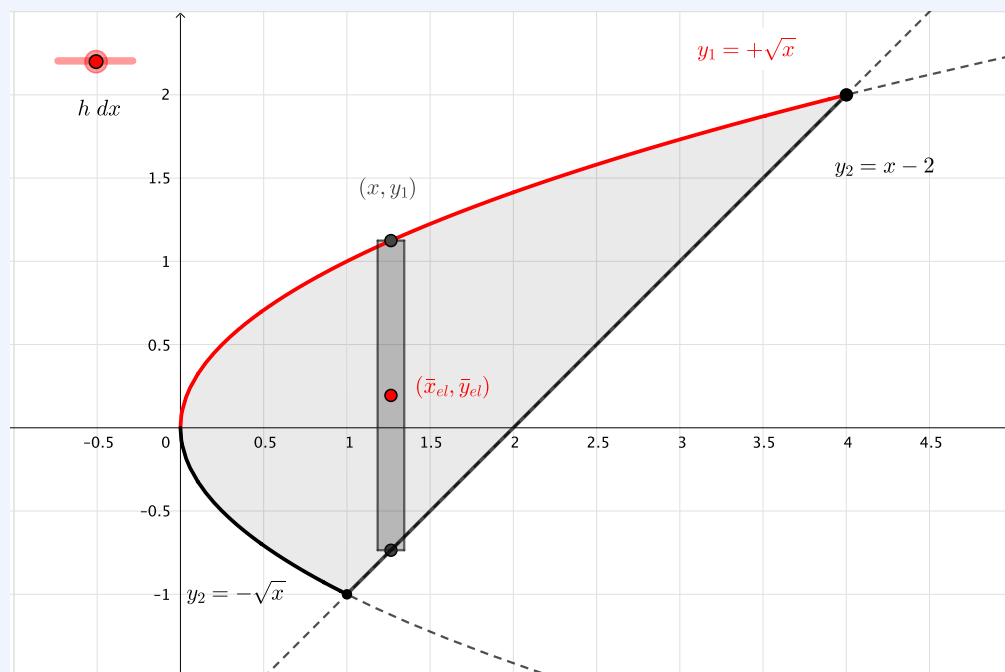
for vertical elements.

With these modifications either strip can be used to find the area and centroid of a shape.



The interactive below compares horizontal and vertical strips for a shape bounded by the parabola $y^2 = x$ and the diagonal line $y = x - 2$. Horizontal strips are a better choice in this case, because the left and right

boundaries are easy to express as functions of y . If vertical strips are chosen, the parabola must be expressed as two different functions of x , and two integrals are needed to cover the area, the first from $x = 0$ to $x = 1$, and the second from $x = 1$ to $x = 4$.



[Live Link](#)

Figure 7.7.5 Function demonstrating good and bad choices of differential elements.

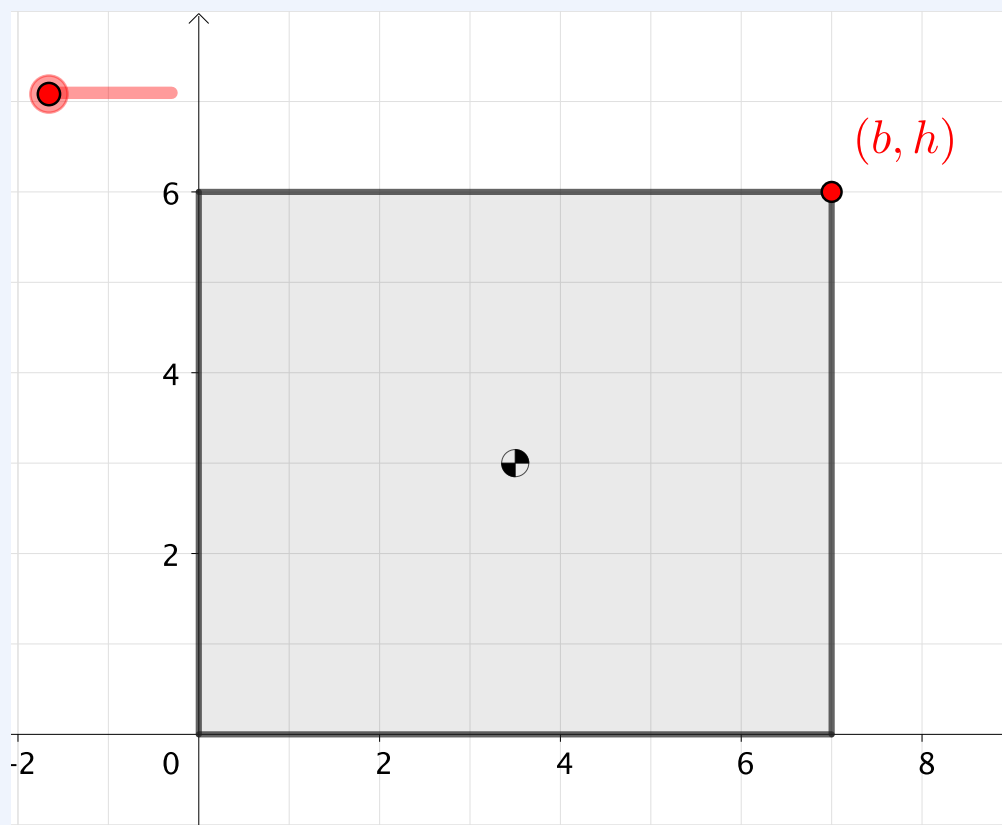
7.7.3 Examples

This section contains several examples of finding centroids by integration, starting with very simple shapes and getting progressively more difficult. All the examples include interactive diagrams to help you visualize the integration process, and to see how dA is related to x or y .

The first two examples are a rectangle and a triangle evaluated three different ways: with vertical strips, horizontal strips, and using double integration. The different approaches produce identical results, as you would expect. You should try to decide which method is easiest for a particular situation.

Example 7.7.6 Centroid of a rectangle.

Use integration to show that the centroid of a rectangle with a base b and a height of h is at its center.



[Live Link](#)

Figure 7.7.7

Solution 1. This solution demonstrates solving integrals using vertical rectangular strips. Set the slider on the diagram to $h \, dx$ to see a representative element.

1. *Set up the integrals.*

The bounding functions in this example are vertical lines $x = 0$ and $x = a$, and horizontal lines $y = 0$ and $y = h$.

The strip extends from $(x, 0)$ on the x axis to (x, h) on the top of the rectangle, and has a differential width dx .

The area of the strip is the base times the height, so

$$dA = h \, dx.$$

The centroid of the strip is located at its midpoint so, by inspection

$$\begin{aligned}\bar{x}_{\text{el}} &= x \\ \bar{y}_{\text{el}} &= h/2\end{aligned}$$

With vertical strips the variable of integration is x , and the limits on x run from $x = 0$ at the left to $x = b$ on the right. For a rectangle, both 0 and h are constants, but in other situations, $\bar{y}_{\text{el}}$ and the left or right limits may be functions of x .

2. *Solve the integrals.*

Substitute dA , $\bar{x}_{\text{el}}$, and $\bar{y}_{\text{el}}$ into (7.7.2) and integrate.

$$\begin{aligned}A &= \int dA & Q_x &= \int \bar{y}_{\text{el}} dA & Q_y &= \int \bar{x}_{\text{el}} dA \\ &= \int_0^b h dx & &= \int_0^b \frac{h}{2} (h dx) & &= \int_0^b x (h dx) \\ &= [hx]_0^b & &= \frac{h^2}{2} \int_0^b dx & &= h \int_0^b x dx \\ &= hb - 0 & &= \frac{h^2}{2} [x]_0^b & &= h \left[\frac{x^2}{2} \right]_0^b \\ A &= bh & Q_x &= \frac{h^2 b}{2} & Q_y &= \frac{b^2 h}{2}\end{aligned}$$

Unsurprisingly, we learn that the area of a rectangle is base times height. Since the area formula is well known, it was not really necessary to solve the first integral. Note that A has units of $[\text{length}]^2$, and Q_x and Q_y have units of $[\text{length}]^3$.

3. *Find the centroid.*

Substituting the results into the definitions gives

$$\begin{aligned}\bar{x} &= \frac{Q_y}{A} & \bar{y} &= \frac{Q_x}{A} \\ &= \frac{b^2 h}{2} \bigg/ bh & &= \frac{h^2 b}{2} \bigg/ bh \\ &= \frac{b}{2} & &= \frac{h}{2}\end{aligned}$$

Solution 2. This solution demonstrates solving integrals using horizontal rectangular strips. Set the slider on the diagram to $b dy$ to see a representative element.

1. *Set up the integrals.*

The bounding functions $x = 0$, $x = a$, $y = 0$ and $y = h$.

The strip extends from $(0, y)$ on the y axis to (b, y) on the right, and has a differential height dy .

The area of the strip is the base times the height, so

$$dA = b \, dy.$$

The centroid of the strip is located at its midpoint so, by inspection

$$\bar{x}_{\text{el}} = b/2$$

$$\bar{y}_{\text{el}} = y$$

With horizontal strips the variable of integration is y , and the limits on y run from $y = 0$ at the bottom to $y = h$ at the top.

For a rectangle, both 0 and h are constants, but in other situations, $\bar{x}_{\text{el}}$ and the upper or lower limits may be functions of y .

2. *Solve the integrals.*

Substitute dA , $\bar{x}_{\text{el}}$, and $\bar{y}_{\text{el}}$ into (7.7.2) and integrate. The results are the same as we found using vertical strips.

$$\begin{aligned} A &= \int dA & Q_x &= \int \bar{y}_{\text{el}} \, dA & Q_y &= \int \bar{x}_{\text{el}} \, dA \\ &= \int_0^h b \, dy & &= \int_0^h y \, (b \, dy) & &= \int_0^h \frac{b}{2} (b \, dy) \\ &= [by]_0^h & &= b \int_0^h y \, dy & &= \frac{b^2}{2} \int_0^h dy \\ &= bh & &= b \left[\frac{y^2}{2} \right]_0^h & &= \frac{b^2}{2} [y]_0^h \\ A &= bh & Q_x &= \frac{h^2 b}{2} & Q_y &= \frac{b^2 h}{2} \end{aligned}$$

3. *Find the centroid.*

Substituting the results into the definitions gives

$$\begin{aligned} \bar{x} &= \frac{Q_y}{A} & \bar{y} &= \frac{Q_x}{A} \\ &= \frac{b^2 h}{2} \bigg/ bh & &= \frac{h^2 b}{2} \bigg/ bh \\ &= \frac{b}{2} & &= \frac{h}{2} \end{aligned}$$

Solution 3. This solution demonstrates solving integrals using square elements and double integrals. Set the slider on the diagram to $dx \, dy$ to see a representative element.

1. *Set up the integrals.*

Set the slider on the diagram to $dx \, dy$ to see a representative element.

The bounding functions $x = 0$, $x = a$, $y = 0$ and $y = h$.

Instead of strips, the integrals will be evaluated using square elements with width dx and height dy located at (x, y) .

The area of the square element is the base times the height, so

$$dA = dx \, dy = dy \, dx.$$

The centroid of the strip is located at its midpoint so, by inspection

$$\bar{x}_{\text{el}} = x$$

$$\bar{y}_{\text{el}} = y$$

We will integrate twice, first with respect to y and then with respect to x . The limits on the first integral are $y = 0$ to h and $x = 0$ to b on the second. For a rectangle, both b and h are constants. In other situations, the upper or lower limits may be functions of x or y .

2. *Solve the integrals.*

Substitute dA , $\bar{x}_{\text{el}}$, and $\bar{y}_{\text{el}}$ into (7.7.2) and integrate the ‘inside’ integral, then the ‘outside’ integral. The results are the same as before.

$$\begin{aligned}
 A &= \int dA & Q_x &= \int \bar{y}_{\text{el}} \, dA & Q_y &= \int \bar{x}_{\text{el}} \, dA \\
 &= \int_0^b \int_0^h dy \, dx & &= \int_0^b \int_0^h y \, dy \, dx & &= \int_0^b \int_0^h x \, dy \, dx \\
 &= \int_0^b \left[\int_0^h dy \right] dx & &= \int_0^b \left[\int_0^h y \, dy \right] dx & &= \int_0^b x \left[\int_0^h dy \right] dx \\
 &= \int_0^b \left[y \right]_0^h dx & &= \int_0^b \left[\frac{y^2}{2} \right]_0^h dx & &= \int_0^b x \left[y \right]_0^h dx \\
 &= h \int_0^b dx & &= \frac{h^2}{2} \int_0^b dx & &= h \int_0^b x \, dx \\
 &= h \left[x \right]_0^b & &= \frac{h^2}{2} \left[x \right]_0^b & &= h \left[\frac{x^2}{2} \right]_0^b \\
 A &= hb & Q_x &= \frac{h^2 b}{2} & Q_y &= \frac{b^2 h}{2}
 \end{aligned}$$

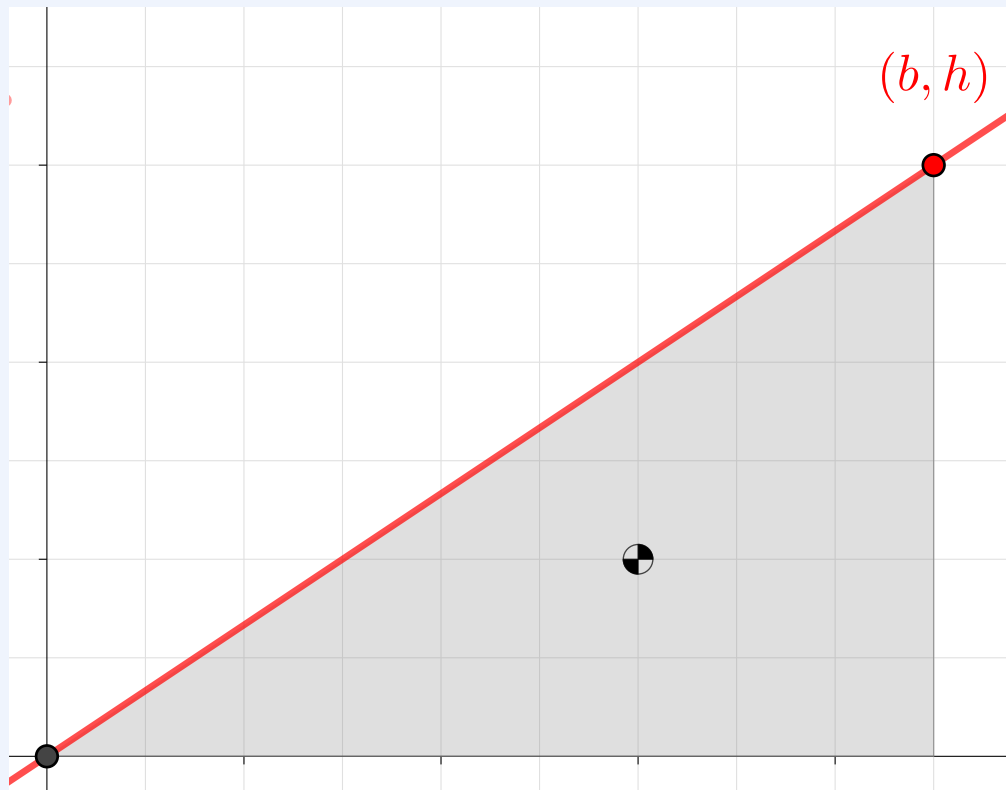
3. Find the centroid.

Substituting the results into the definitions gives

$$\begin{aligned}\bar{x} &= \frac{Q_y}{A} & \bar{y} &= \frac{Q_x}{A} \\ &= \frac{b^2 h}{2} \Big/ bh & &= \frac{h^2 b}{2} \Big/ bh \\ &= \frac{b}{2} & &= \frac{h}{2}\end{aligned}$$

Example 7.7.8 Centroid of a triangle.

Use integration to locate the centroid of a triangle with base b and height of h oriented as shown in the interactive.



[Live Link](#)

Figure 7.7.9

Solution 1. This solution demonstrates finding the centroid of the triangle using vertical strips $dA = y \, dx$. Set the slider on the diagram to $y \, dx$ to see a representative element.

1. *Set up the integrals.*

The bounding functions in this example are the x axis, the vertical line $x = b$, and the straight line through the origin with a slope of $\frac{h}{b}$. Using the slope-intercept form of the equation of a line, the upper bounding function is

$$y = f(x) = \frac{h}{b}x$$

and any point on this line is designated (x, y) .

The strip extends from $(x, 0)$ on the x axis to (x, y) on the function, has a height of y , and a differential width dx . The area of this strip is

$$dA = y \, dx.$$

The centroid of the strip is located at its midpoint so, by inspection

$$\begin{aligned}\bar{x}_{\text{el}} &= x \\ \bar{y}_{\text{el}} &= y/2\end{aligned}$$

With vertical strips the variable of integration is x , and the limits are $x = 0$ to $x = b$.

2. *Solve the integrals.*

Substitute dA , $\bar{x}_{\text{el}}$, and $\bar{y}_{\text{el}}$ into (7.7.2) and integrate. In contrast to the rectangle example both dA and $\bar{y}_{\text{el}}$ are functions of x , and will have to be integrated accordingly.

$$\begin{aligned}A &= \int dA & Q_x &= \int \bar{y}_{\text{el}} \, dA & Q_y &= \int \bar{x}_{\text{el}} \, dA \\ &= \int_0^b y \, dx & &= \int_0^b \frac{y}{2} (y \, dx) & &= \int_0^b x (y \, dx) \\ &= \int_0^b \frac{h}{b} x \, dx & &= \frac{1}{2} \int_0^b \left(\frac{h}{b} x\right)^2 dx & &= \int_0^b x \left(\frac{h}{b} x\right) dx \\ &= \frac{h}{b} \left[\frac{x^2}{2}\right]_0^b & &= \frac{h^2}{2b^2} \int_0^b x^2 dx & &= \frac{h}{b} \int_0^b x^2 dx \\ &= \frac{h}{b} \frac{b^2}{2} & &= \frac{h^2}{2b^2} \left[\frac{x^3}{3}\right]_0^b & &= \frac{h}{b} \left[\frac{x^3}{3}\right]_0^b \\ A &= \frac{bh}{2} & Q_x &= \frac{h^2 b}{6} & Q_y &= \frac{b^2 h}{3}\end{aligned}$$

We learn that the area of a triangle is one half base times height. Since the area formula is well known, it would have been more efficient to skip the first integral. Note that A has units of $[\text{length}]^2$, and Q_x and Q_y have units of $[\text{length}]^3$.

3. *Find the Centroid.*

Substituting the results into the definitions gives

$$\begin{aligned}\bar{x} &= \frac{Q_y}{A} & \bar{y} &= \frac{Q_x}{A} \\ &= \frac{b^2h}{3} \bigg/ \frac{bh}{2} & &= \frac{h^2b}{6} \bigg/ \frac{bh}{2} \\ &= \frac{2}{3}b & &= \frac{1}{3}h\end{aligned}$$

.

Solution 2. This solution demonstrates solving integrals using horizontal rectangular strips. Set the slider on the diagram to $(b - x) dy$ to see a representative element.

1. *Set up the integrals.*

As before, the triangle is bounded by the x axis, the vertical line $x = b$, and the line

$$y = f(x) = \frac{h}{b}x.$$

To integrate using horizontal strips, the function $f(x)$ must be inverted to express x in terms of y . Solving for $f(x)$ for x gives

$$x = g(y) = \frac{b}{h}y.$$

The limits on the integral are from $y = 0$ to $y = h$.

The strip extends from (x, y) to (b, y) , has a height of dy , and a length of $(b - x)$, therefore the area of this strip is

$$dA = (b - x)dy.$$

The coordinates of the midpoint of the element are

$$\begin{aligned}\bar{y}_{\text{el}} &= y \\ \bar{x}_{\text{el}} &= x + \frac{(b - x)}{2} = \frac{b + x}{2}\end{aligned}$$

.

These expressions are recognized as the *average* of the x and y coordinates of strip's endpoints.

A common student mistake is to use $dA = x \, dy$, and $\bar{x}_{\text{el}} = x/2$. These would be correct if you were looking for the properties of the area to the left of the curve.

2. *Solve the integrals.*

Substitute dA , $\bar{x}_{\text{el}}$, and $\bar{y}_{\text{el}}$ into the definitions of Q_x and Q_y and integrate. The results are the same as we found using vertical strips. There is no need to evaluate $A = \int dA$ since we know that $A = \frac{bh}{2}$ for a triangle.

$$\begin{aligned}
 Q_x &= \int \bar{y}_{\text{el}} \, dA & Q_y &= \int \bar{x}_{\text{el}} \, dA \\
 &= \int_0^h y (b-x) \, dy & &= \int_0^h \frac{(b+x)}{2} (b-x) \, dy \\
 &= \int_0^h (by - xy) \, dy & &= \frac{1}{2} \int_0^h (b^2 - x^2) \, dy \\
 &= \int_0^h \left(by - \frac{by^2}{h} \right) dy & &= \frac{1}{2} \int_0^h \left(b^2 - \frac{b^2 y^2}{h^2} \right) dy \\
 &= b \left[\frac{y^2}{2} - \frac{y^3}{3h} \right]_0^h & &= \frac{b^2}{2} \left[y - \frac{y^3}{3h^2} \right]_0^h \\
 &= bh^2 \left(\frac{1}{2} - \frac{1}{3} \right) & &= \frac{1}{2} (b^2 h) \left(1 - \frac{1}{3} \right) \\
 Q_x &= \frac{h^2 b}{6} & Q_y &= \frac{b^2 h}{3}
 \end{aligned}$$

It makes solving these integrals easier if you avoid prematurely substituting in the function for x and if you factor out constants whenever possible. Here it $x = g(y)$ was not substituted until the fourth line.

3. *Find the centroid.*

Substituting the results into the definitions gives

$$\begin{aligned}
 \bar{x} &= \frac{Q_y}{A} & \bar{y} &= \frac{Q_x}{A} \\
 &= \frac{b^2 h}{3} \bigg/ \frac{bh}{2} & &= \frac{h^2 b}{6} \bigg/ \frac{bh}{2} \\
 &= \frac{2}{3} b & &= \frac{1}{3} h
 \end{aligned}$$

Solution 3. This solution demonstrates solving integrals using square elements and double integrals. Set the slider on the diagram to $dx \, dy$ or $dy \, dx$ to see a representative element.

1. *Set up the integrals.*

As before, the triangle is bounded by the x axis, the vertical line $x = b$, and the line

$$y = f(x) = \frac{h}{b}x \quad \text{or in terms of } y, \quad x = g(y) = \frac{b}{h}y.$$

In this solution the integrals will be evaluated using square differential elements $dA = dy \, dx$ located at (x, y) .

With double integration, you must take care to evaluate the limits correctly, since the limits on the inside integral are functions of the variable of integration of the outside integral. The inside integral essentially stacks the elements into strips and the outside integral adds all the strips to cover the area.

Choosing to express dA as $dy \, dx$ means that the integral over y will be conducted first. The limits on the inside integral are from $y = 0$ to $y = f(x)$. Then, the limits on the outside integral are from $x = 0$ to $x = b$.

Using $dA = dx \, dy$ would reverse the order of integration, so the inside integral's limits would be from $x = g(y)$ to $x = b$, and the limits on the outside integral would be $y = 0$ to $y = h$. Either choice will give the same results — *if you don't make any errors!*

The area of the square element is the base times the height, so

$$dA = dy \, dx.$$

The centroid of the square is located at its midpoint so, by inspection

$$\bar{x}_{\text{el}} = x$$

$$\bar{y}_{\text{el}} = y$$

2. *Solve the integrals.*

Substitute dA , $\bar{x}_{\text{el}}$, and $\bar{y}_{\text{el}}$ into (7.7.2) and integrate the ‘inside’ integral, then the ‘outside’ integral. The results are the same as before.

$$Q_x = \int \bar{y}_{\text{el}} \, dA \qquad Q_y = \int \bar{x}_{\text{el}} \, dA$$

$$\begin{aligned}
&= \int_0^b \int_0^{f(x)} y \, dy \, dx &= \int_0^b \int_0^{f(x)} x \, dy \, dx \\
&= \int_0^b \left[\int_0^{f(x)} y \, dy \right] dx &= \int_0^b x \left[\int_0^{f(x)} dy \right] dx \\
&= \int_0^b \left[\frac{y^2}{2} \right]_0^{f(x)} dx &= \int_0^b x \left[y \right]_0^{f(x)} dx \\
&= \frac{1}{2} \int_0^b \left[\frac{h^2}{b^2} x^2 \right] dx &= \int_0^b x \left[\frac{h}{b} x \right] dx \\
&= \frac{h^2}{2b^2} \int_0^b x^2 dx &= \frac{h}{b} \int_0^b x^2 dx \\
&= \frac{h^2}{2b^2} \left[\frac{x^3}{3} \right]_0^b &= \frac{h}{b} \left[\frac{x^3}{3} \right]_0^b \\
Q_x &= \frac{h^2 b}{6} &Q_y &= \frac{b^2 h}{3}
\end{aligned}$$

3. Find the centroid.

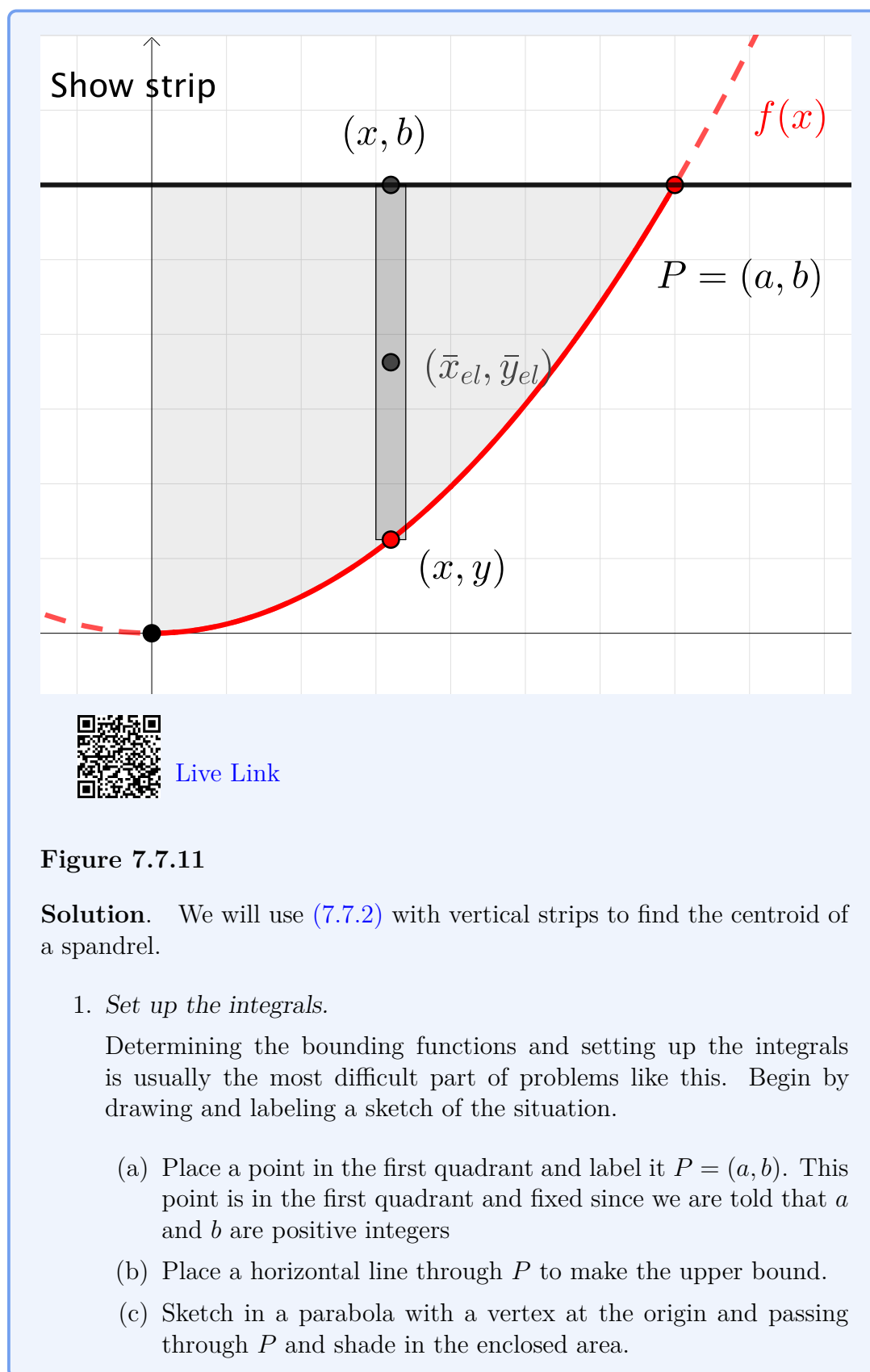
Substituting Q_x and Q_y along with $A = bh/2$ into the centroid definitions gives

$$\begin{aligned}
\bar{x} &= \frac{Q_y}{A} & \bar{y} &= \frac{Q_x}{A} \\
&= \frac{b^2 h}{3} \bigg/ \frac{bh}{2} & &= \frac{h^2 b}{6} \bigg/ \frac{bh}{2} \\
&= \frac{2}{3} b & &= \frac{1}{3} h
\end{aligned}$$

The next two examples involve areas with functions for both boundaries,

Example 7.7.10 Centroid of a semi-parabola.

Find the coordinates of the centroid of a parabolic spandrel bounded by the y axis, a horizontal line passing through the point (a, b) , and a parabola with a vertex at the origin and passing through the same point. a and b are positive integers.



- (d) Decide which differential element you intend to use. For this example we choose to use vertical strips, which you can see if you tick **show strips** in the interactive above. Horizontal strips $dA = x \, dy$ would give the same result, but you would need to define the equation for the parabola in terms of y .

Determining the equation of the parabola and expressing it in terms of x and any known constants is a critical step. You should remember from algebra that the general equation of parabola with a vertex at the origin is $y = kx^2$, where k is a constant which determines the shape of the parabola. If $k > 0$, the parabola opens upward and if $k < 0$, the parabola opens downward.

To find the value of k , substitute the coordinates of P into the general equation, then solve for k .

$$\begin{aligned} y &= kx^2, \text{ so at } P \\ (b) &= k(a)^2 \\ k &= \frac{b}{a^2} \end{aligned}$$

The resulting function of the parabola is

$$y = y(x) = \frac{b}{a^2}x^2.$$

To perform the integrations, express the area and centroidal coordinates of the element in terms of the points at the top and bottom of the strip. The area of the strip is its height times its base, so

$$dA = (b - y) \, dx.$$

If you incorrectly used $dA = y \, dx$, you would find the centroid of the spandrel below the curve.

For vertical strips, the bottom is at (x, y) on the parabola, and the top is directly above at (x, b) . The strip has a differential width dx . The centroid of the strip is located at its midpoint and the coordinates are found by averaging the x and y coordinates of the points at the top and bottom.

$$\begin{aligned} \bar{x}_{el} &= (x + x)/2 = x \\ \bar{y}_{el} &= (y + b)/2 \end{aligned}$$

For vertical strips, the integrations are with respect to x , and the limits on the integrals are $x = 0$ on the left to $x = a$ on the right.

2. Solve the integrals.

We have already established that $y(x) = kx^2$ where $k = b/a^2$. To simplify the algebra, it is best not to prematurely substitute $y(x)$ and k , but you must substitute in any functions of x before you do the integration step.

$$\begin{aligned}
 A &= \int dA & Q_x &= \int \bar{y}_{el} dA & Q_y &= \int \bar{x}_{el} dA \\
 &= \int_0^a (b - y) dx & &= \int_0^a \frac{(b + y)}{2} (b - y) dx & &= \int_0^a x(b - y) dx \\
 &= \int_0^a (b - kx^2) dx & &= \frac{1}{2} \int_0^a (b^2 - y^2) dx & &= \int_0^a x(b - y) dx \\
 &= bx - k \frac{x^3}{3} \Big|_0^a & &= \frac{1}{2} \int_0^a (b^2 - (kx^2)^2) dx & &= \int_0^a x(b - kx^2) dx \\
 &= ba - k \frac{a^3}{3} & &= \frac{1}{2} \int_0^a (b^2 - k^2 x^4) dx & &= \int_0^a (bx - kx^3) dx \\
 &= ba - \left(\frac{b}{a^2} \right) \frac{a^3}{3} & &= \frac{1}{2} \left[b^2 x - k^2 \frac{x^5}{5} \right]_0^a & &= \left[\frac{bx^2}{2} - k \frac{x^4}{4} \right]_0^a \\
 &= \frac{3ba}{3} - \frac{ba}{3} & &= \frac{1}{2} \left[b^2 a - \left(\frac{b}{a^2} \right)^2 \frac{a^5}{5} \right] & &= \left[\frac{ba^2}{2} - \left(\frac{b}{a^2} \right) \frac{a^4}{4} \right] \\
 &= \frac{2}{3} ba & &= \frac{1}{2} b^2 a \left[1 - \frac{1}{5} \right] & &= ba^2 \left[\frac{1}{2} - \frac{1}{4} \right] \\
 A &= \frac{2}{3} ba & Q_x &= \frac{2}{5} b^2 a & Q_y &= \frac{1}{4} ba^2
 \end{aligned}$$

The area of the spandrel is $2/3$ of the area of the enclosing rectangle and the moments of area have units of $[\text{length}]^3$.

3. Find the centroid.

Substituting the results into the definitions gives

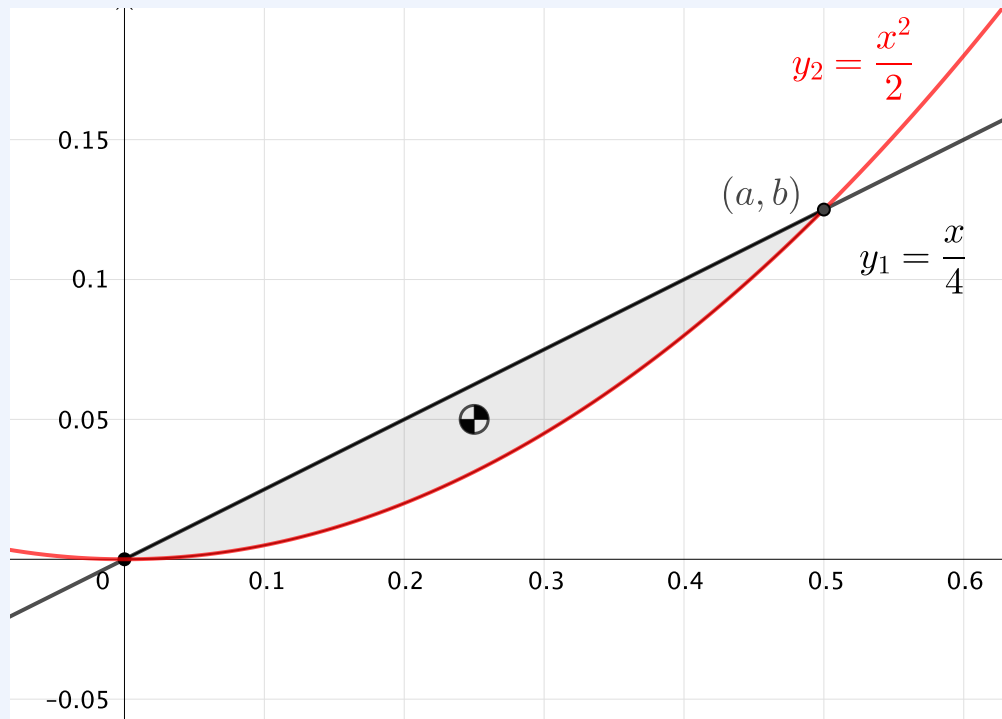
$$\begin{aligned}
 \bar{x} &= \frac{Q_y}{A} & \bar{y} &= \frac{Q_x}{A} \\
 &= \frac{ba^2}{4} \Big/ \frac{2ba}{3} & &= \frac{2b^2 a}{5} \Big/ \frac{2ba}{3} \\
 &= \frac{3}{8} a & &= \frac{3}{5} b
 \end{aligned}$$

$\bar{x}$ is $3/8$ of the width and $\bar{y}$ is $3/5$ of the height of the enclosing rectangle.

Example 7.7.12 Centroid of an area between two curves.

Use integration to locate the centroid of the area bounded by

$$y_1 = \frac{x}{4} \text{ and } y_2 = \frac{x^2}{2}.$$



[Live Link](#)

Figure 7.7.13

Find the centroid location $(\bar{x}, \bar{y})$ of the shaded area between the two curves below.

Solution 1. This solution demonstrates finding the centroid of the area between two functions using vertical strips $dA = y \, dx$. Set the slider on the diagram to $h \, dx$ to see a representative element.

1. *Set up the integrals.*

The bounding functions in this example are the x axis, the vertical line $x = b$, and the straight line through the origin with a slope of $\frac{h}{b}$. Using the slope-intercept form of the equation of a line, the upper bounding function is

$$y = f(x) = \frac{h}{b}x$$

and any point on this line is designated (x, y) .

The strip extends from $(x, 0)$ on the x axis to (x, y) on the function, has a height of y , and a differential width dx . The area of this strip is

$$dA = ydx.$$

The centroid of the strip is located at its midpoint so, by inspection

$$\bar{x}_{\text{el}} = x$$

$$\bar{y}_{\text{el}} = y/2$$

With vertical strips the variable of integration is x , and the limits are $x = 0$ to $x = b$.

2. Solve the integrals.

Substitute dA , $\bar{x}_{\text{el}}$, and $\bar{y}_{\text{el}}$ into (7.7.2) and integrate. In contrast to the rectangle example both dA and $\bar{y}_{\text{el}}$ are functions of x , and will have to be integrated accordingly.

$$\begin{aligned} A &= \int dA \\ &= \int_0^{1/2} (y_1 - y_2) dx \\ &= \int_0^{1/2} \left(\frac{x}{4} - \frac{x^2}{2} \right) dx \\ &= \left[\frac{x^2}{8} - \frac{x^3}{6} \right]_0^{1/2} \\ &= \left[\frac{1}{32} - \frac{1}{48} \right] \\ A &= \frac{1}{96} \end{aligned}$$

$$\begin{aligned} Q_x &= \int \bar{y}_{\text{el}} dA & Q_y &= \int \bar{x}_{\text{el}} dA \\ &= \int_0^{1/2} \left(\frac{y_1 + y_2}{2} \right) (y_1 - y_2) dx & &= \int_0^{1/2} x(y_1 - y_2) dx \\ &= \frac{1}{2} \int_0^{1/2} (y_1^2 - y_2^2) dx & &= \int_0^{1/2} x \left(\frac{x}{4} - \frac{x^2}{2} \right) dx \\ &= \frac{1}{2} \int_0^{1/2} \left(\frac{x^2}{16} - \frac{x^4}{4} \right) dx & &= \int_0^{1/2} \left(\frac{x^2}{4} - \frac{x^3}{2} \right) dx \\ &= \frac{1}{2} \left[\frac{x^3}{48} - \frac{x^5}{20} \right]_0^{1/2} & &= \left[\frac{x^3}{12} - \frac{x^4}{8} \right]_0^{1/2} \end{aligned}$$

$$Q_x = \frac{1}{1920} \quad Q_y = \frac{1}{384}$$

3. *Find the Centroid.*

Substituting the results into the definitions gives

$$\begin{aligned} \bar{x} &= \frac{Q_y}{A} & \bar{y} &= \frac{Q_x}{A} \\ &= \frac{1}{384} \bigg/ \frac{1}{96} & &= \frac{1}{1920} \bigg/ \frac{1}{96} \\ \bar{x} &= \frac{1}{4} & \bar{y} &= \frac{1}{20} \end{aligned}$$

Solution 2. This solution demonstrates finding the centroid of the area between two functions using vertical strips $dA = y \, dx$. Set the slider on the diagram to $h \, dx$ to see a representative element.

1. *Set up the integrals.*

The bounding functions in this example are the x axis, the vertical line $x = b$, and the straight line through the origin with a slope of $\frac{h}{b}$. Using the slope-intercept form of the equation of a line, the upper bounding function is

$$y = f(x) = \frac{h}{b}x$$

and any point on this line is designated (x, y) .

The strip extends from $(x, 0)$ on the x axis to (x, y) on the function, has a height of y , and a differential width dx . The area of this strip is

$$dA = ydx.$$

The centroid of the strip is located at its midpoint so, by inspection

$$\begin{aligned} \bar{x}_{\text{el}} &= x \\ \bar{y}_{\text{el}} &= y/2 \end{aligned}$$

With vertical strips the variable of integration is x , and the limits are $x = 0$ to $x = b$.

2. Solve the integrals.

Substitute dA , $\bar{x}_{\text{el}}$, and $\bar{y}_{\text{el}}$ into (7.7.2) and integrate. In contrast to the rectangle example both dA and $\bar{y}_{\text{el}}$ are functions of x , and will have to be integrated accordingly.

$$\begin{aligned}
 A &= \int dA \\
 &= \int_0^y (x_2 - x_1) dy \\
 &= \int_0^{1/8} (4y - \sqrt{2y}) dy \\
 &= \left[2y^2 - \frac{4}{3}y^{3/2} \right]_0^{1/8} \\
 &= \left[\frac{1}{32} - \frac{1}{48} \right] \\
 A &= \frac{1}{96}
 \end{aligned}$$

$$\begin{aligned}
 Q_x &= \int \bar{y}_{\text{el}} dA & Q_y &= \int \bar{x}_{\text{el}} dA \\
 &= \int_0^{1/8} y(x_2 - x_1) dy & &= \int_0^{1/8} \left(\frac{x_2 + x_1}{2} \right) (x_2 - x_1) dy \\
 &= \int_0^{1/8} y(\sqrt{2y} - 4y) dy & &= \frac{1}{2} \int_0^{1/8} (x_2^2 - x_1^2) dy \\
 &= \int_0^{1/8} (\sqrt{2}y^{3/2} - 4y^2) dy & &= \frac{1}{2} \int_0^{1/8} (2y - 16y^2) dy \\
 &= \left[\frac{2\sqrt{2}}{5}y^{5/2} - \frac{4}{3}y^3 \right]_0^{1/8} & &= \frac{1}{2} \left[y^2 - \frac{16}{3}y^3 \right]_0^{1/8} \\
 &= \left[\frac{1}{320} - \frac{1}{384} \right] & &= \frac{1}{2} \left[\frac{1}{64} - \frac{1}{96} \right] \\
 Q_x &= \frac{1}{1920} & Q_y &= \frac{1}{384}
 \end{aligned}$$

3. Find the Centroid.

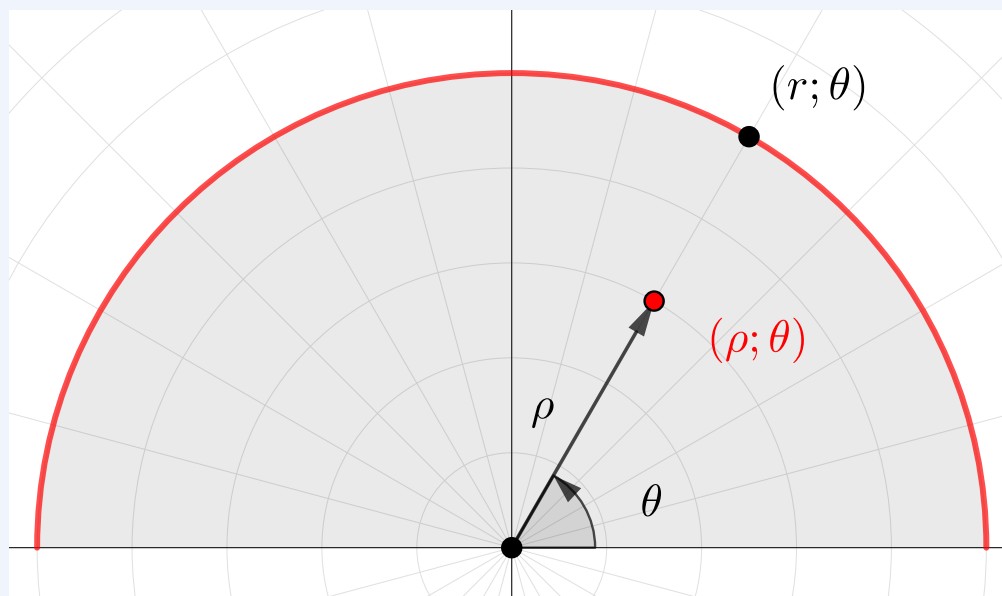
Substituting the results into the definitions gives

$$\begin{aligned}
 \bar{x} &= \frac{Q_y}{A} & \bar{y} &= \frac{Q_x}{A} \\
 &= \frac{1}{384} \bigg/ \frac{1}{96} & &= \frac{1}{1920} \bigg/ \frac{1}{96} \\
 \bar{x} &= \frac{1}{4} & \bar{y} &= \frac{1}{20}
 \end{aligned}$$

The last example demonstrates using double integration with polar coordinates.

Example 7.7.14 Centroid of a semi-circle.

Find the coordinates of the top half of a circle with radius r , centered at the origin.



[Live Link](#)

Figure 7.7.15

Solution. We will use (7.7.2) with polar coordinates (ρ, θ) to solve this problem because they are a natural fit for the geometry. In polar coordinates, the equation for the bounding semicircle is simply

$$\rho = r.$$

Normally this involves evaluating three integrals but as you will see, we can take some shortcuts in this problem. Otherwise we will follow the same procedure as before.

1. *Set up the integrals.*

Divide the semi-circle into "rectangular" differential elements of area dA , as shown in the interactive when you select **Show element**. This shape is not really a rectangle, but in the limit as $d\rho$ and $d\theta$ approach zero, it doesn't make any difference.

The radial height of the rectangle is $d\rho$ and the tangential width is the arc length $\rho d\theta$. The product is the differential area dA .

$$dA = (d\rho)(\rho d\theta) = \rho d\rho d\theta. \quad (7.7.3)$$

The differential element is located at (ρ, θ) in polar coordinates. Expressing this point in rectangular coordinates gives

$$\bar{x}_{\text{el}} = \rho \cos \theta$$

$$\bar{y}_{\text{el}} = \rho \sin \theta$$

.

2. *Solve the integrals.*

The area of a semicircle is well known, so there is no need to actually evaluate $A = \int dA$,

$$A = \int dA = \frac{\pi r^2}{2}.$$

Since the semi-circle is symmetrical about the y axis,

$$Q_y = \int \bar{x}_{\text{el}} dA = 0.$$

This is because each element of area to the right of the y axis is balanced by a corresponding element the same distance the left which cancel each other out in the sum.

All that remains is to evaluate the integral Q_x in the numerator of

$$\bar{y} = \frac{Q_x}{A} = \frac{\bar{y}_{\text{el}} dA}{A}$$

The differential area dA is the product of two differential quantities, we will need to perform a double integration.

$$\begin{aligned} Q_x &= \int \bar{y}_{\text{el}} dA \\ &= \int_0^\pi \int_0^r (\rho \sin \theta) \rho d\rho d\theta \\ &= \int_0^\pi \sin \theta \left[\int_0^r \rho^2 d\rho \right] d\theta \\ &= \int_0^\pi \sin \theta \left[\frac{\rho^3}{3} \right]_0^r d\theta \\ &= \frac{r^3}{3} \int_0^\pi \sin \theta d\theta \end{aligned}$$

$$\begin{aligned}
 &= \frac{r^3}{3} [-\cos \theta]_0^\pi \\
 &= -\frac{r^3}{3} [\cos \pi - \cos 0] \\
 &= -\frac{r^3}{3} [(-1) - (1)] \\
 Q_x &= \frac{2}{3}r^3
 \end{aligned}$$

3. Find the centroid.

Substituting the results into the definitions gives

$$\begin{aligned}
 \bar{y} &= \frac{Q_x}{A} \\
 &= \frac{2r^3}{3} \bigg/ \frac{\pi r^2}{2} \\
 &= \frac{4r}{3\pi}
 \end{aligned}$$

So $\bar{x} = 0$ and lies on the axis of symmetry, and $\bar{y} = \frac{4r}{3\pi}$ above the diameter.

This result can be extended by noting that a semi-circle is mirrored quarter-circles on either side of the y axis. These must have the same $\bar{y}$ value as the semi-circle. Further, quarter-circles are symmetric about a 45° line, so for the quarter-circle in the first quadrant,

$$\bar{x} = \bar{y} = \frac{4r}{3\pi}.$$

7.8 Distributed Loads

Key Questions

- What is a distributed load?
- Given a distributed load, how do we find the magnitude of the equivalent concentrated force?
- Given a distributed load, how do we find the location of the equivalent concentrated force?

Distributed loads are forces which are spread out over a length, area, or

volume. Most real-world loads are distributed, including the weight of building materials and the force of wind, water, or earth pushing on a surface. Pressure, load, weight density and stress are all names commonly used for distributed loads. Distributed load is a force per unit length or force per unit area depicted with a series of force vectors joined together at the top, and will be designated as $w(x)$ to indicate that the distributed loading is a function of x .

For example, although a shelf of books *could* be treated as a collection of individual forces, it is more common and convenient to represent the weight of the books as a **uniformly distributed load**. A uniformly distributed load is a load which has the same value everywhere, i.e. $w(x) = C$, a constant.



Figure 7.8.1 A shelf of books with various weights.

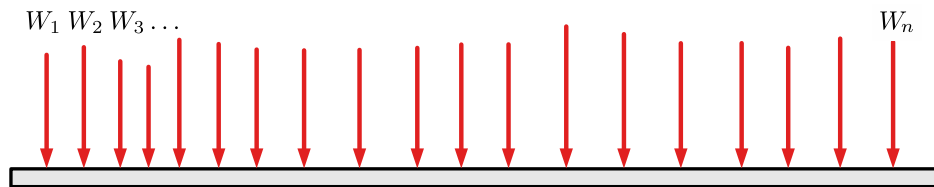


Figure 7.8.2 Each book represented as an individual weight

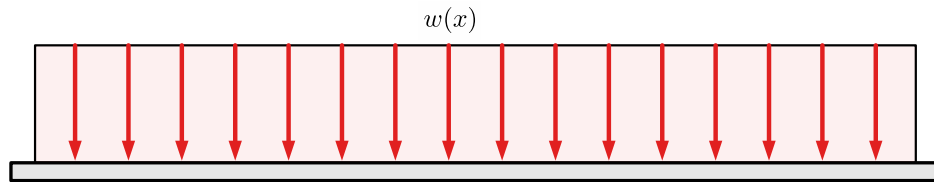


Figure 7.8.3 All the books represented as a distributed load.

We can use the computational tools discussed in the previous chapters to handle distributed loads if we first convert them to equivalent point forces. This equivalent replacement must be the **resultant** of the distributed loading, as discussed in [Section 4.7](#). Recall that this resultant force has the same external effect on the object as the original system of forces did.

To be equivalent, the point force must have a:

- Magnitude equal to the area or volume under the distributed load function.
- Line of action that passes through the centroid of the distributed load

distribution.

The next two sections will explore how to find the magnitude and location of the equivalent point force for a distributed load.

7.8.1 Equivalent Magnitude

The magnitude of the distributed load of the books is the total weight of the books divided by the length of the shelf

$$w(x) = \frac{\sum W_i}{\ell}.$$

It represents the *average* book weight per unit length. Similarly, the total weight of the books is equal to the value of the distributed load times the length of the shelf or

$$W = w(x)\ell$$

$$\text{total weight} = \frac{\text{weight}}{\text{length}} \times \text{length of shelf}$$

This total load is simply the area under the curve $w(x)$, and has units of force. If the loading function is not uniform, integration may be necessary to find the area.

Example 7.8.4 Bookshelf.

A common paperback is about 3 cm thick and weighs approximately 3 N. What is the loading function $w(x)$ for a shelf full of paperbacks and what is the total weight of paperback books on a 6 m shelf?

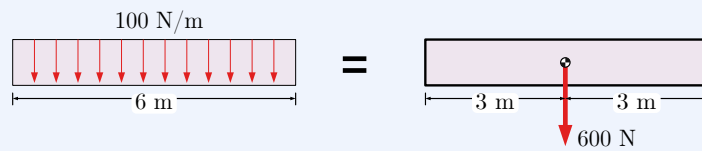
Solution. The weight of one paperback over its thickness is the load intensity $w(x)$, so

$$w(x) = \frac{3 \text{ N}}{3 \text{ cm}} = 100 \text{ N/m}.$$

The total weight is the area under the load intensity diagram, which in this case is a rectangle. So, a 6 m bookshelf covered with paperbacks would have to support

$$W = w(x)\ell = (100 \text{ N/m})(6 \text{ m}) = 600 \text{ N}.$$

The line of action of this equivalent load passes through the centroid of the rectangular loading, so it acts at $x = 3 \text{ m}$.



7.8.2 Equivalent Location

To use a distributed load in an equilibrium problem, you must know the equivalent magnitude to sum the forces, and also know the position or line of action to sum the moments.

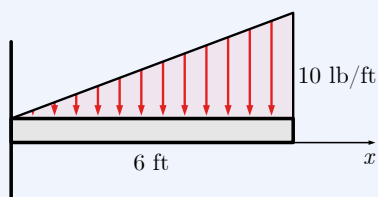
The line of action of the equivalent force acts through the centroid of area under the load intensity curve. For a rectangular loading, the centroid is in the center. We know the vertical and horizontal coordinates of this centroid, but since the equivalent point force's line of action is vertical and we can slide a force along its line of action, the vertical coordinate of the centroid is not important in this context.

Similarly, for a triangular distributed load — also called a **uniformly varying load** — the magnitude of the equivalent force is the area of the triangle, $bh/2$ and the line of action passes through the centroid of the triangle. The horizontal distance from the larger end of the triangle to the centroid is $\bar{x} = b/3$.

Essentially, we're finding the balance point so that the moment of the force to the left of the centroid is the same as the moment of the force to the right.

The examples below will illustrate how you can combine the computation of both the magnitude and location of the equivalent point force for a series of distributed loads.

Example 7.8.5 Uniformly Varying Load.



Find the equivalent point force and its point of application for the distributed load shown.

Solution. The equivalent load is the ‘area’ under the triangular load intensity curve and it acts straight down at the centroid of the triangle. This triangular loading has a 6 ft base and a 10 lb/ft height so

$$W = \frac{1}{2}bh = \frac{1}{2}(6 \text{ ft})(10 \text{ lb/ft}) = 30 \text{ lb.}$$

and the centroid is located $2/3$ of the way from the left end so,

$$\bar{x} = 4 \text{ ft.}$$

Distributed loads may be any geometric shape or defined by a mathematical function. If the load is a combination of common shapes, use the properties of the shapes to find the magnitude and location of the equivalent point force using the methods of [Section 7.5](#). If the distributed load is defined by a mathematical function, integrate to find their area using the methods of [Section 7.7](#).

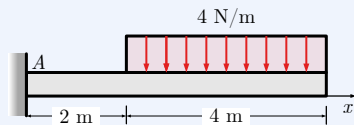
A few things to note:

- You can include the distributed load or the equivalent point force on your free-body diagram, *but not both!*
- Since you're calculating an area, you can divide the area up into any shapes you find convenient. So, if you don't recall the area of a trapezoid off the top of your head, break it up into a rectangle and a triangle.

7.8.3 Distributed Load Applications

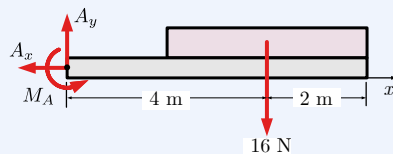
Once you convert distributed loads to the resultant point force, you can solve problem in the same manner that you have other problems in previous chapters of this book. Note that while the resultant forces are *externally* equivalent to the distributed loads, they are not *internally* equivalent, as will be shown [Chapter 8](#).

Example 7.8.6 Cantilever Beam.



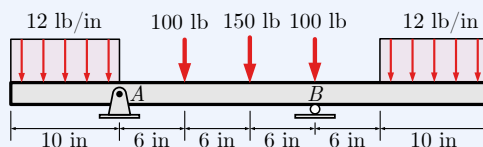
Find the reactions at the fixed connection at A.

Solution. Draw a free-body diagram with the distributed load replaced with an equivalent concentrated load, then apply the equations of equilibrium.



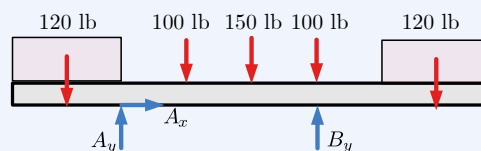
$$\begin{aligned}\Sigma F_x = 0 &\rightarrow A_x = 0 \\ \Sigma F_y = 0 &\rightarrow A_y = 16 \text{ N} \\ \Sigma M_A = 0 &\rightarrow M_A = (16 \text{ N})(4 \text{ m}) \\ &= 64 \text{ N}\cdot\text{m}\end{aligned}$$

Example 7.8.7 Beam Reactions.



Find the reactions at the supports for the beam shown.

Solution. Start by drawing a free-body diagram of the beam with the two distributed loads replaced with equivalent concentrated loads. The two distributed loads are $(10 \text{ in})(12 \text{ lb/in}) = 120 \text{ lb}$ each.



Then apply the equations of equilibrium.

$$\begin{aligned} \sum M_A &= 0 \\ &+(12 \text{ lb/in})(10 \text{ in})(5 \text{ in}) - (100 \text{ lb})(6 \text{ in}) \\ &\quad - (150 \text{ lb})(12 \text{ in}) - (100 \text{ lb})(18 \text{ in}) \\ &+(B_y)(18 \text{ in}) - (12 \text{ lb/in})(10 \text{ in})(29 \text{ in}) = 0 \rightarrow B_y = 393.3 \text{ lb} \end{aligned}$$

$$\begin{aligned} \sum F_y &= 0 \\ &-(12 \text{ lb/in})(10 \text{ in}) + A_y - 100 \text{ lb} - 150 \text{ lb} \\ &\quad - 100 \text{ lb} + B_y - (12 \text{ lb/in})(10 \text{ in}) = 0 \rightarrow A_y = 196.7 \text{ lb} \end{aligned}$$

$$\sum F_x = 0 \rightarrow A_x = 0$$

7.9 Fluid Statics

Key Questions

- What is the basic relationship between depth and pressure?
- How are absolute and relative pressure different?
- How can use our knowledge of centroids to compute the equivalent point forces of fluids?

Pressure is the term used for a force distributed over an area

$$P = \frac{F}{A}. \quad (7.9.1)$$

We will consider the effect of fluid pressure on underwater surfaces, including slanted or curved objects. In all cases we will simply ask the question: what is the pressure at each point and how does it change along the surface?

Pressure can be measured in two different ways

- **Absolute pressure** is the pressure measured above an absolute or perfect vacuum. The absolute pressure of the surrounding atmosphere is approximately 101.3 kPa or 14.7 lb/in², and a perfect vacuum is 0 psi or 0 kPa.
- **Gage pressure** is the pressure indicated by a standard pressure gage. The gage reads zero when exposed directly to the atmosphere, positive when the pressure is higher than atmospheric pressure, and negative pressure indicates a vacuum. In effect, pressure gages ignores the pressure of the atmosphere which surrounds us.

We will use gage pressure for the remainder of the chapter. Commonly used pressure units include:

- 1 pascals (Pa) = 1 N/m²
- 1 kilopascal (kPa) = 1000 N/m²
- 1 pound per square inch (psi) = lb/in²
- 1 kip per square inch (ksi) = 1000 lb/in²
- 1 pound per square foot (psf) = 1 lb/ft²

7.9.1 Principles of Fluid Statics

A fluid, like water or air exerts a pressure on its surroundings. This pressure applies a distributed load on surfaces surrounding the fluid, like the face of a dam, an irrigation control gate, a teakettle, or the drum of a steam boiler.

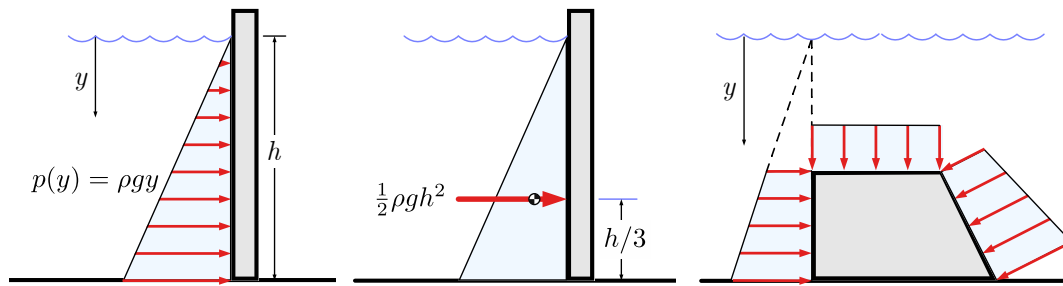
When you dive underwater, the pressure you feel in your ears increases with depth. At the surface, the gage pressure is zero no matter which unit system you use. As you descend, the fluid pressure P increases with depth according to the equation

$$P = \rho gh, \quad (7.9.2)$$

where:

- ρ is the density of a fluid,
- g is gravitational acceleration, and
- h is the height of fluid above the point of interest.

Since fluid pressure increases linearly with depth, it behaves as a distributed load which increases linearly from 0 at the surface to ρgh at depth h , acting normal to the surface. The pressure can be replaced with an equivalent force acting through the centroid of the triangular loading, with a magnitude equal to the triangular area. The pressure on horizontal surfaces is constant, and it is normal to all surfaces.



(a) Distributed pressure. (b) Equivalent force. (c) Pressure is perpendicular to the surface.

Figure 7.9.1 Pressure on submerged surfaces.

Some points to remember when solving fluid pressure problems.

1. The pressure due to the fluid always acts perpendicular the surface.
2. A particle underwater will feel the same pressure from all directions.
3. Pressure increases linearly with depth. $P = \rho gh$
4. $P = \rho gh$ assumes a constant density and thus is valid only for incompressible fluids like water or oil, but not for compressible fluids like air.
5. In English units, specific weight γ is often used instead of density ρ to describe fluids. Specific weight is the *weight* per unit volume of a substance, while density is its *mass* per unit volume. The two properties are related by $\gamma = \rho g$. The specific weight of freshwater at room temperature is about 62.4 lb/ft^3 .
6. Gage pressure is the pressure above the surrounding atmospheric pressure. Atmospheric pressure is approximately 14.7 lb/in^2 or 101.3 kPa , but since this pressure acts on everything equally and from all directions, the pressure scale can be offset to make the pressure of the surroundings 0 lb/in^2 , gage.

Question 7.9.2

Does fluid pressure depend on the surface area of the container? For instance, is the pressure below the Atlantic Ocean less than the pressure below the Pacific Ocean since the Pacific is larger?

Answer. No. Fluid pressure is a function of density and depth only, so the surface area of an ocean or tank is insignificant.

$$P = \rho gh.$$

Assuming that the density of seawater and g are the same everywhere under the ocean, the gage pressure depends on depth only.

Question 7.9.3

Compare the pressure at three feet and thirty feet below the surface of freshwater to the atmospheric pressure.

Answer. The gage pressure at 3 ft is

$$p = \gamma d = 62.4 \text{ lb/ft}^3 \times 3 \text{ ft} = 187 \text{ lb/ft}^2 = 1.30 \text{ lb/in}^2.$$

This is

$$\frac{14.7 \text{ lb/in}^2 + 1.3 \text{ lb/in}^2}{14.7 \text{ lb/in}^2} = 1.088,$$

approximately 9% greater than atmospheric pressure.

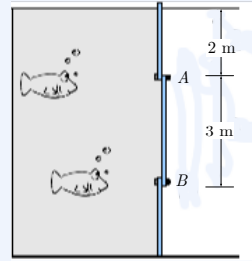
At 30 ft below the surface, the pressure is 10 times higher, 13.0 lb/in² which is nearly twice atmospheric pressure.

7.9.2 Fluid Statics Applications

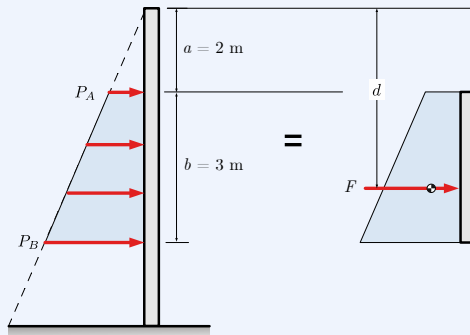
Example 7.9.4 Force on a submerged window.

An aquarium tank has a 3 m × 1.5 m window AB for viewing the inhabitants. The tank contains water with density $\rho = 1000 \text{ kg/m}^3$.

Find the force of the water on the window, and the location of the equivalent point load.



Solution. Begin by drawing a diagram of the window showing the load intensity and the equivalent concentrated force.



The pressure at the top and the bottom of the window are

$$P_A = \rho g(2 \text{ m}) = 19620 \text{ N/m}^2$$

$$P_B = \rho g(5 \text{ m}) = 49050 \text{ N/m}^2$$

Since the loading is linear, the average pressure acting on the window is

$$\begin{aligned} P_{ave} &= (P_A + P_B)/2 \\ &= 34300 \text{ N/m}^2 \end{aligned}$$

The total force acting on the window is the average pressure times the area of the window

$$\begin{aligned} F &= (P_{ave})(3 \text{ m} \times 1.5 \text{ m}) \\ &= 155 \text{ kN} \end{aligned}$$

This force may also be visualized as the volume of a trapezoidal prism with a 1.5 m depth into the page.

The line of action of the equivalent force passes through the centroid of the trapezoid, which may be calculated using composite areas, see [Section 7.5](#). Dividing the trapezoid into a triangle and a rectangle and measuring down from the surface of the tank, the distance to the equivalent force is

$$\begin{aligned} d &= \frac{\sum A_i \bar{y}_i}{\sum A_i} \\ d &= \frac{[P_A(3 \text{ m})](3.5 \text{ m}) + \left[\frac{1}{2}(P_B - P_A)(3 \text{ m}) \right](4 \text{ m})}{[P_A(3 \text{ m})] + \left[\frac{1}{2}(P_B - P_A)(3 \text{ m}) \right]} \\ d &= 3.71 \text{ m} \end{aligned}$$

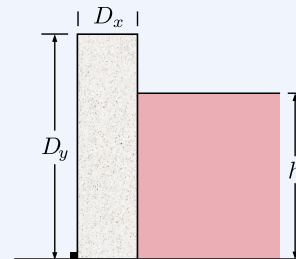
If you prefer, you may use the formula from the [Centroid table](#) to locate the centroid of the trapezoid instead.

Example 7.9.5 Mud on Concrete Wall.

A concrete retaining wall 3 m tall and 0.8 m thick encloses wet clay mud.

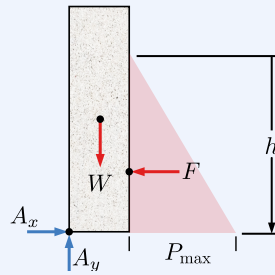
Find the maximum depth h of mud that can be retained without tipping the wall.

Assume the density of mud is 1760 kg/m^3 and the density of concrete is 2400 kg/m^3 .



Solution. Begin by drawing free-body diagram of the wall, showing the forces acting on it: the weight of the concrete W_c , acting at the center of gravity of the rectangular wall, the horizontal force F due to the mud, and the reaction components at A . Define symbol D_z for the unspecified

length of the wall into the page.



The weight of the concrete is found by multiplying the weight density of concrete $\lambda_c = \rho_c g$ by the volume of the rectangular wall.

$$\begin{aligned} W_c &= \lambda_c V \\ &= (\rho_c g) (D_x D_y D_z) \end{aligned}$$

The horizontal force is caused by the hydraulic pressure of the mud on the submerged face of the wall. The pressure is a distributed load that increases linearly from zero at the surface to $P_{\max} = \rho_m g h$ at depth h . Since the loading is triangular, the average pressure is one-half the maximum pressure, and the resulting force is:

$$\begin{aligned} F &= P_{\text{ave}} A = \frac{1}{2} P_{\max} A \\ &= \frac{1}{2} (\rho_m g h) (h D_z) \end{aligned}$$

The wall will tip if the counter-clockwise moment due to F is greater than the clockwise moment caused by the concrete's weight. By taking moments about point A , we can find the maximum allowable height of the mud, h . Note that the length of the dam and g appear on both sides and cancel out.

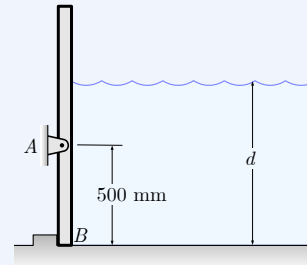
$$\begin{aligned} \Sigma M_A &= 0 \\ (W_c) \left(\frac{D_x}{2} \right) &= F \left(\frac{h}{3} \right) \\ (\rho_c \cancel{g} D_x D_y \cancel{D_z}) \left(\frac{D_x}{2} \right) &= \left(\frac{1}{2} \rho_m \cancel{g} h^2 \cancel{D_z} \right) \left(\frac{h}{3} \right) \\ h^3 &= 3 \frac{\rho_c}{\rho_m} D_x^2 D_y \\ h^3 &= 3 \frac{2400 \text{ kg/m}^3}{1760 \text{ kg/m}^3} (0.8 \text{ m})^2 (3.0 \text{ m}) \\ h &= 1.98 \text{ m} \end{aligned}$$

Example 7.9.6 Sea Gate.

A sea gate is hinged at point A and is designed to rotate and release the water when the depth d exceeds a certain value.

The gate extends 2 m into the page. The mass density of the water is $\rho = 1000 \text{ kg/m}^3$.

What depth will cause the gate to open?



Solution. For the gate to tip, the equivalent force of the water must act at or above A . That happens when the centroid of the load intensity diagram of the water is at or above A , so

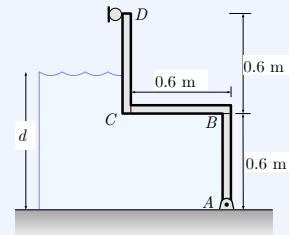
$$\frac{d}{3} \geq 500 \text{ mm}$$

$$d \geq 1500 \text{ mm}$$

Example 7.9.7 Gate with Horizontal Surface.

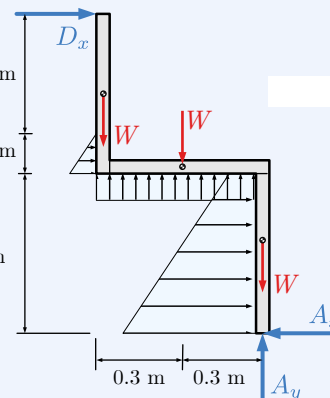
A gate at the end of a freshwater channel is fabricated from three 125 kg, $0.6 \text{ m} \times 1 \text{ m}$ rectangular steel plates. The gate is hinged at A and rests against a frictionless support at D . The depth of the water d is 0.75 m.

Draw the free-body diagram and determine the reactions at A and D .

**Solution.**

A free-body diagram of a cross section of the gate is shown. For simplicity the thickness of the steel plates has been ignored. You should ensure that sufficient distances are provided to locate the loads.

The easiest way to solve this is to apply the principle of transmissibility: slide the lower trapezoid left until it aligns with the upper triangle and makes a triangular loading.



The total horizontal force from the water will be

$$F_x = P_{ave} A$$

$$\begin{aligned}
&= \left[\frac{1}{2} \rho g (0.75 \text{ m}) \right] (0.75 \text{ m} \times 1 \text{ m}) \\
&= \left[\frac{1}{2} (1000 \text{ kg/m}^3) (9.81 \text{ m/s}^2) (0.75 \text{ m}) \right] (0.75 \text{ m} \times 1 \text{ m}) \\
&= 2760 \text{ N}
\end{aligned}$$

acting to the right 0.25 m above point A .

The total vertical load from the water is

$$\begin{aligned}
F_y &= P_{ave} A \\
&= [\rho g (0.15 \text{ m})] (0.6 \text{ m} \times 1 \text{ m}) \\
&= [(1000 \text{ kg/m}^3) (9.81 \text{ m/s}^2) (0.15 \text{ m})] (0.6 \text{ m} \times 1 \text{ m}) \\
&= 882.9 \text{ N}
\end{aligned}$$

acting upward 0.3 m to the left of A .

Each plate weighs

$$\begin{aligned}
W &= mg \\
&= (125 \text{ kg}) (9.81 \text{ m/s}^2) \\
&= 1226 \text{ N}
\end{aligned}$$

From here solve the equilibrium equations to find the reactions. You should complete this for practice.

$$\begin{array}{lll}
\Sigma M_A = 0 & \implies & D_x = 239 \text{ N} \rightarrow \\
\Sigma F_x = 0 & \implies & A_x = 2998 \text{ N} \leftarrow \\
\Sigma F_y = 0 & \implies & A_y = 2795 \text{ N} \uparrow
\end{array}$$

7.10 Exercises (Ch. 7)

Chapter 7 Exercises

Centroids by Composite Area	0/110
Centroid of parts	0/30
	Not attempted
Shape with Hole	0/40
	Not attempted
Rectangles and Triangles	0/40
	Not attempted
Centroids by Integration	0/130
Differential Elements	0/30
	Not attempted
Area under a curve	0/20
	Not attempted
Centroids by integration	0/60
	Not attempted
Upper and lower bounds are functions	0/20
	Not attempted
Distributed Loads	0/60
Uniformly distributed load	0/20
	Not attempted
Uniformly varying load	0/20
	Not attempted
Trapezoidal loading	0/20
	Not attempted

Display options

Done

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Standalone
In Context